

Emotion

情绪和文化实验室 | 浙江大学

https://zju-eclab.github.io/

Outline

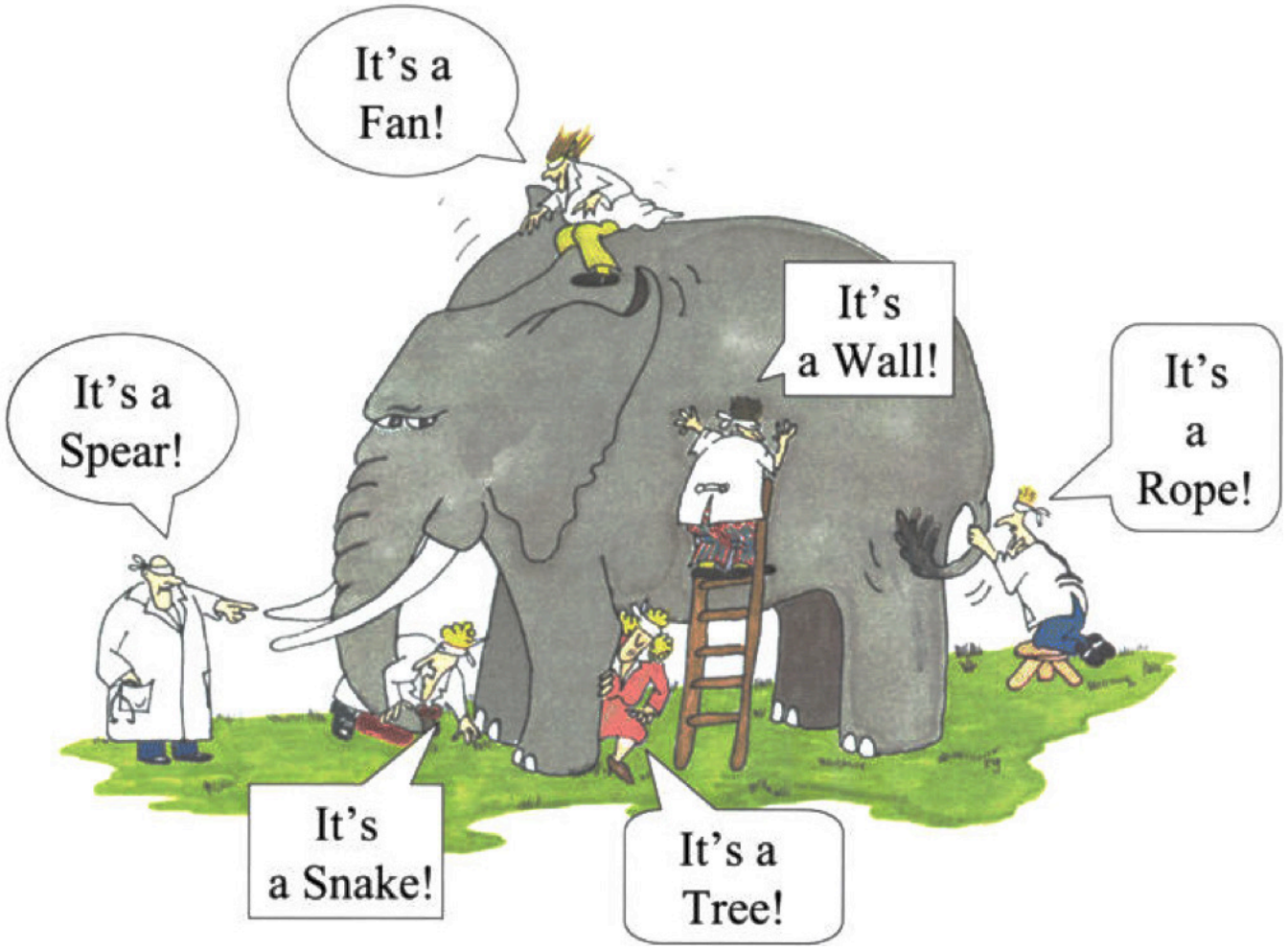
1) Characterizing Emotion
2) Theories of emotion:
    - Evolutionary
    - Appraisal
    - Constructionist
3) How to Study Emotions?
    - Inducing Emotions
    - Measuring Emotions
4) Facial Expressions of Emotion

Defining Emotion: Is It Possible?

Everybody knows what an emotion is, until asked to give a definition

Fehr and Russell, 1984, p.464

Defining Emotion



Defining Emotion: Your list

Write down a list of emotions (1 minute)

Defining Emotion

Episodic, relatively short-term, biologically-based patterns of perception, experience, physiology, action, and communication that occur in response to specific physical and social challenges and opportunities.

Keltner and Gross, 1999

情绪是在应对特定身体与社会挑战或机会时出现的，片段性的、相对短期的、以生物学为基础的感知、体验、生理反应、行动和沟通模式。

Defining Emotion: Your List

- Evaluate your list of emotions against Keltner and Gross' definition

Episodic, relatively short-term, biologically-based patterns of perception, experience, physiology, action, and communication that occur in response to specific physical and social challenges and opportunities.

情绪是在应对特定身体与社会挑战或机会时出现的，片段性的、相对短期的、以生物学为基础的感知、体验、生理反应、行动和沟通模式。

Emotion Words

- Anger, Disgust, Fear, Surprise, Sadness, Joy
- Contempt

- 《礼记·礼运》：喜、怒、哀、惧、爱、恶、欲

Emotion Words

第一阶段：情绪词汇的收集与翻译

为了收集完整的情绪词汇列表，我们对2021年以前东西方文化中关于情绪的研究进行了全面回顾。检索关键词包括“情绪、情绪词、面部表情、面孔、数据库、文化”等，覆盖EBSCO、PsychINFO、Google Scholar、Web of Science和CNKI等多个数据库。通过这些检索，收集了大量的情绪词汇和面部表情相关的文献综述、元分析和研究。

此外，我们还将中国日常生活中常见的情绪词汇纳入列表，包括微信表情符号和6名中国本土研究助理的自我报告，以确保所收集的情绪词汇能够反映当代中国文化的情感表达。初步文献检索共识别出579个情绪词汇，补充步骤仅增加了19个新词。

为了确保英中文翻译的准确性，使用了标准的翻译/回译程序。最终成功整理出**598**个中文情绪词汇。

Emotion Words

第一阶段：评估情绪词的典型性与相似性

为了进一步完善情绪词列表，两位研究助理对589个情绪词进行了严格评估，筛选出典型性高、相似性低的词汇。最终，经过筛选，确定了159个情绪词汇。

为进一步精炼，招募了61名参与者评估词对的相似性。159个情绪词可形成12,561个词对，考虑到大多数词对相似性较低（如“幸福”和“尴尬”），3名研究助理筛选出可能引起混淆的174个词对（如“骄傲”和“自豪”，“焦虑”和“不安”），生成四个子集用于评估。



Emotion Words

99个情绪词汇

https://osf.io/57tvb/?view_only=0bb1eb0767f24900b4b234e98f0564fc

28 Chinese Dynamic Facial Expressions of Emotion

Exp1_Evaluation of 99 Emotion Words....

ExpressionChannelSelectonRate

Show rows with cells including:

Emotion	Valence	Unnamed: 3	Arousal	Unnamed: 5	Expressio…	Unnamed: 7	Unnamed: 8	Unnamed: 9	Unnamed: …	Unnamed: …
	M	SD	M	SD	Voice	Facial Expre…	Body Move…	Verbal Lang…	Touch	Art
wei mi (萎靡)	-3.2162162…	0.75037528…	3.48648648…	1.53878779…	0.14	0.86	0.62	0.32	0.05	0.16
jie tuo (解脱)	1.43243243…	1.50075056…	5	1.95789002…	0.22	0.65	0.32	0.49	0.22	0.14
lang man (…	2.40540540…	0.92674638…	6.24324324…	1.46069628…	0.43	0.68	0.62	0.76	0.54	0.43
jian xiao (奸…	-1.5135135…	1.57447736…	5.24324324…	1.27813678…	0.65	0.89	0.27	0.03	0.03	0.03
kong ju (恐惧)	-2.7027027…	1.52506584…	7	1.31233464…	0.81	0.76	0.68	0.43	0.32	0.11
ling wu (领悟)	1.59459459…	1.16570101…	5.18918918…	2.01160297…	0.27	0.59	0.24	0.49	0.05	0.14
hai xiu (害羞)	-0.1621621…	1.01416395…	5.27027027…	1.55721704…	0.14	0.92	0.54	0.22	0.11	0.05
tiao xin (挑衅)	-1.9459459…	1.24601466…	5.94594594…	1.85511224…	0.57	0.73	0.78	0.62	0.35	0
ren jun bu ji…	1.72972972…	1.23936920…	5.78378378…	1.37709709…	0.49	0.95	0.41	0.43	0.11	0
jing wei (敬畏)	1.78378378…	1.10893179…	4.83783783…	1.69170383…	0.16	0.54	0.49	0.49	0.08	0.16
tong qing (…	1	1.15470053…	5.05405405…	1.73118369…	0.11	0.7	0.32	0.7	0.57	0.03
pei fu (佩服)	1.64864864…	1.25202538…	5.35135135…	1.33783025…	0.27	0.59	0.54	0.65	0.35	0.03
meng huan …	1.75675675…	0.98334478…	5.16216216…	1.80298384…	0.27	0.49	0.27	0.46	0.11	0.41
kong huang …	-2.4864864…	1.21613279…	6.43243243…	1.69214755…	0.43	0.76	0.57	0.57	0.22	0.05
qi nei (气馁)	-2.4324324…	1.09393633…	4.37837837…	1.60470404…	0.19	0.89	0.46	0.54	0.08	0.03

28 Chinese Dynamic Facial Expressions of Emotion

Exp1_Emotion Word List.xlsx

LiteratureReview | 598 Words | 159 Words | 99 words

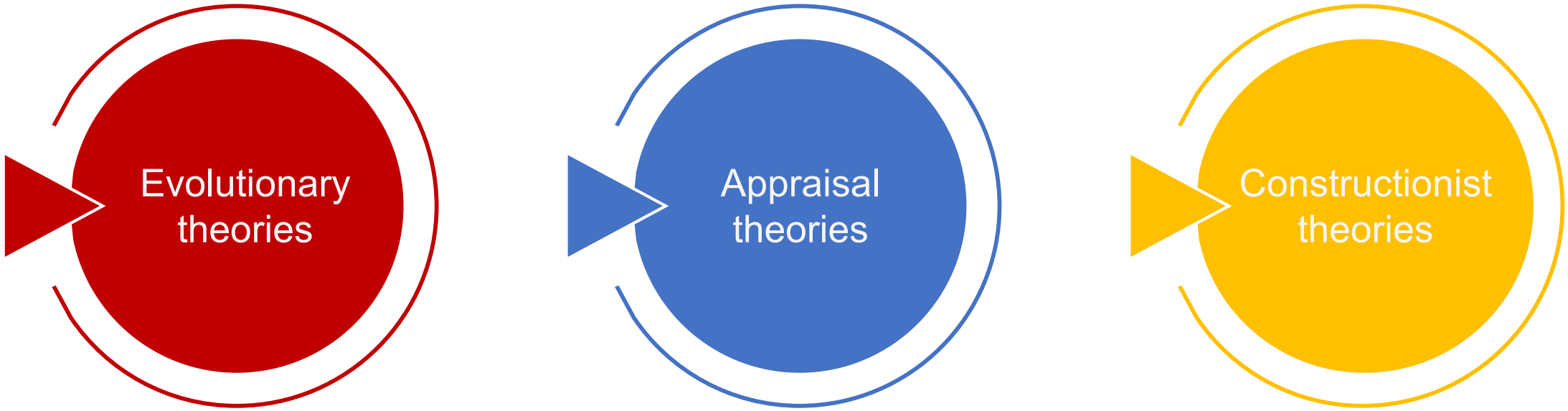
Show rows with cells including:

ID	Reference	Emotion Words
1	Cordaro et al., (2018)	amusement, awe, contentment, coyness, desire, happin…
2	Elfenbein et al., (2007)	anger, contempt, disgust, embarrassment, fear, happines…
3	Tsai et al., (2002)	happiness, pride, love, anger, disgust, sadness
4	Matsumoto & Willingham, (2006)	smile, contempt, disgust, fear, sadness
5	Tracy & Matsumoto, (2008)	pride, shame
6	Soto et al., (2005)	amusement, anxiety, contempt, contentment, disgust, e…
7	Cordaro et al., (2020)	anger, disgust, fear, happiness, sadness, surprise, amus…
8	Cowen et al., (2021)	elation, confusion, pain, concentration, disappointment, i…
9	Volynets et al., (2020)	anger, fear, disgust, happiness, sadness, surprise, anxiet…

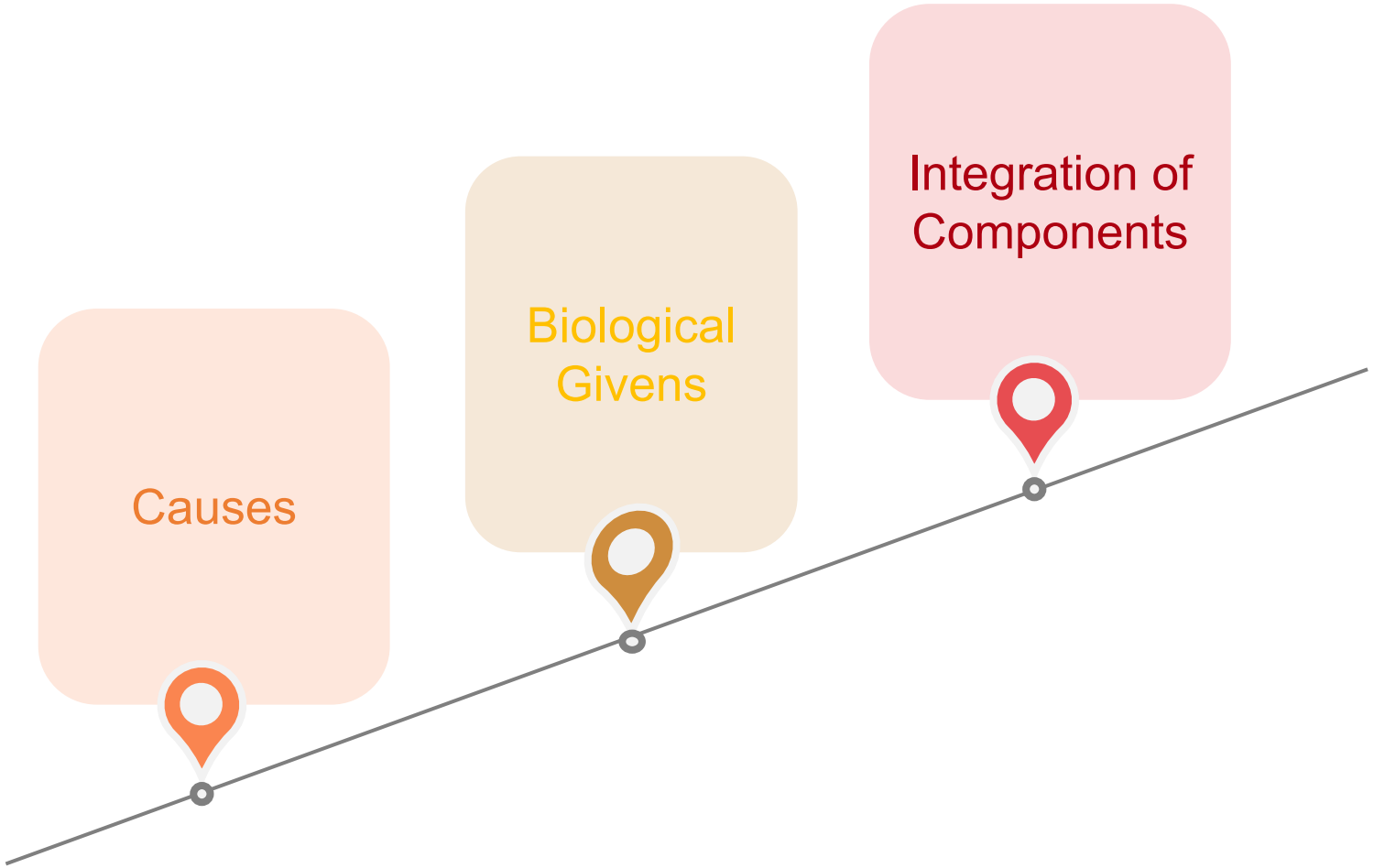
What is a Theory of Emotion?

- Testable statement about exactly how emotions come about.

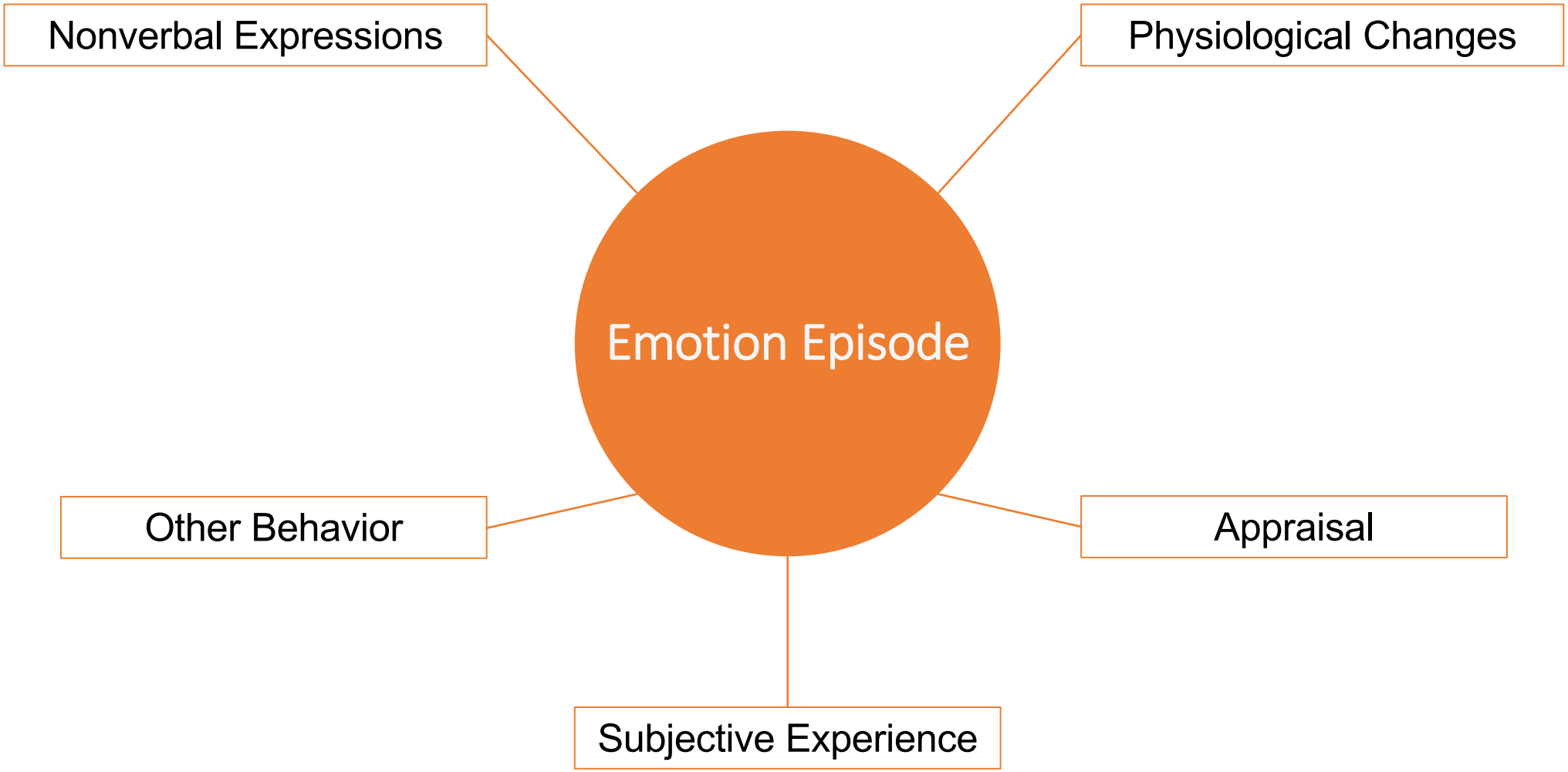
Theories of Emotion



Theories of Emotion



Emotion Components



Evolutionary Theories

Darwin (1872/1998) :

- Facial expressions came about because they were serviceable habits that solved whatever problem elicited in the first place.





Darwin, C. (1872/1998). *The Expression of the Emotions in Man and Animals*. Oxford University Press.

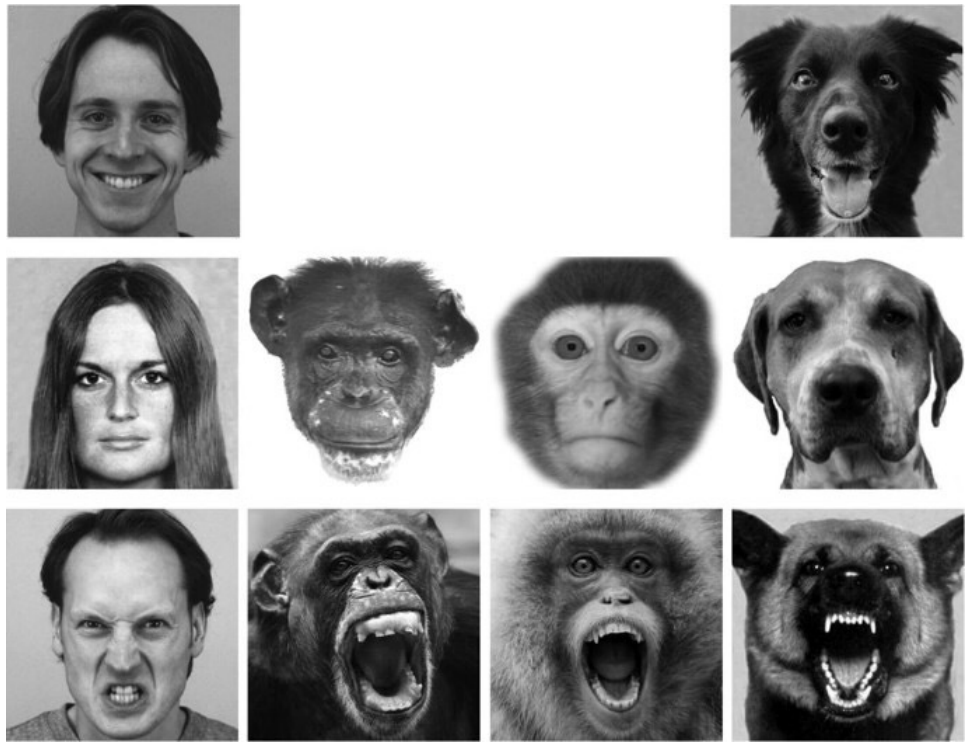
Evolutionary Theories



Evolutionary Theories

- Across culture and species





Evolutionary Theories: Causes of Emotion

- Adaptive problems (Challenges to survival and opportunities for reproduction) are the causes of emotions.
- Emotions result in actions designed by evolution to solve the problem.

Evolutionary Theories: Causes of Emotion

- Emotions are associated with action tendencies.

Table 1.1 Plutchik's theory of actions (behaviors) that respond successfully to fundamental life challenges and opportunities and the emotions that motivate the actions

Emotions and Adaptive Problems			
Adaptive Problem	*Emotion*	*Behavior*	*Outcome*
Threat Obstacle Potential mate Loss of valued person Group member Gruesome object New territory Sudden novel object			

Evolutionary Theories of Emotion

- Evolved and Adaptive
- Separate
- Distinct physiology
- Universal
- Innate



Evolutionary Theories of Emotion

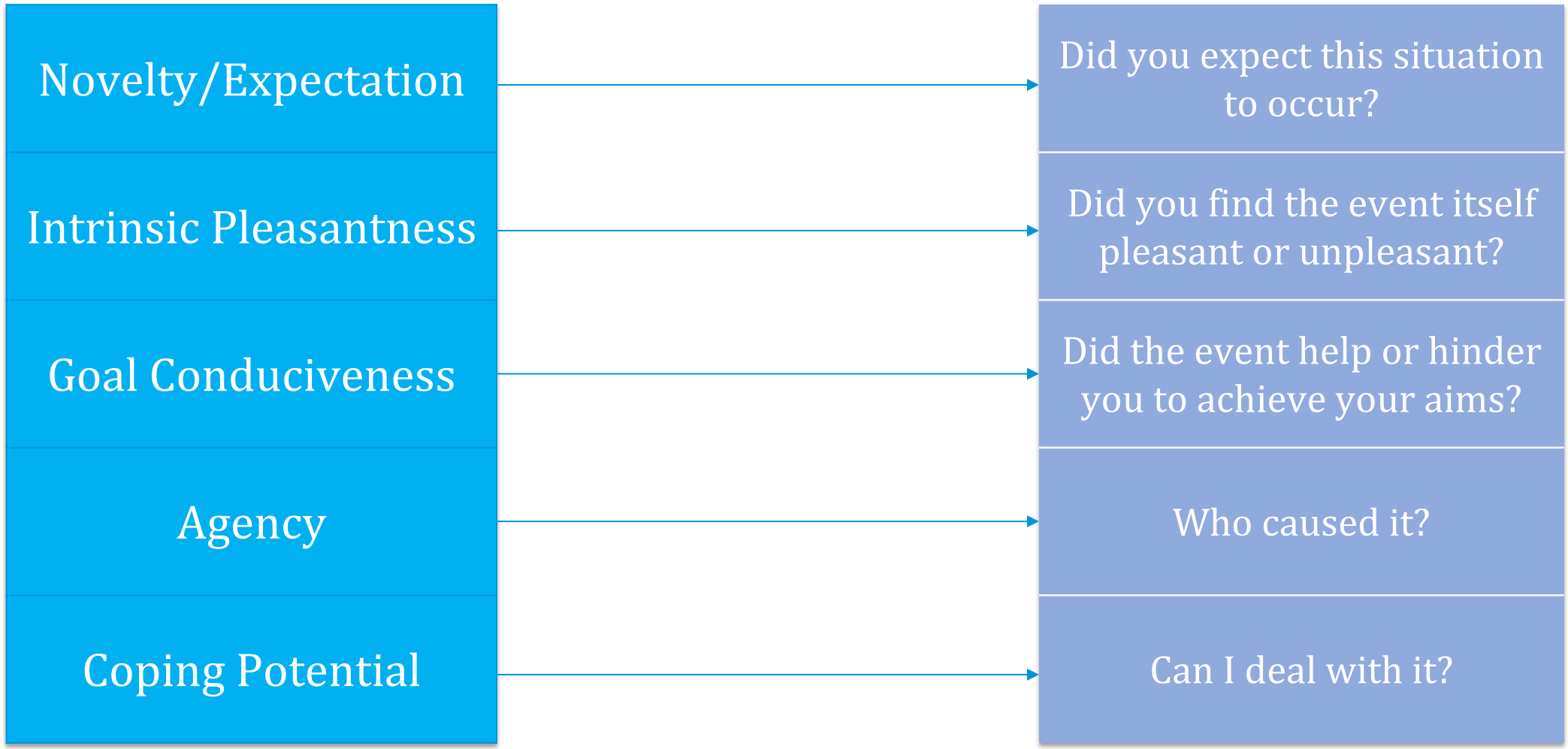
- **Causes:** Emotions caused by activation of Affect Program
- **Biological givens:** Basic emotions are biological givens
- **Components:** Facial expressions, behavior (action tendencies), physiology, changes to cognition; Subjective experience less central component
- **Integration:** High integration between components



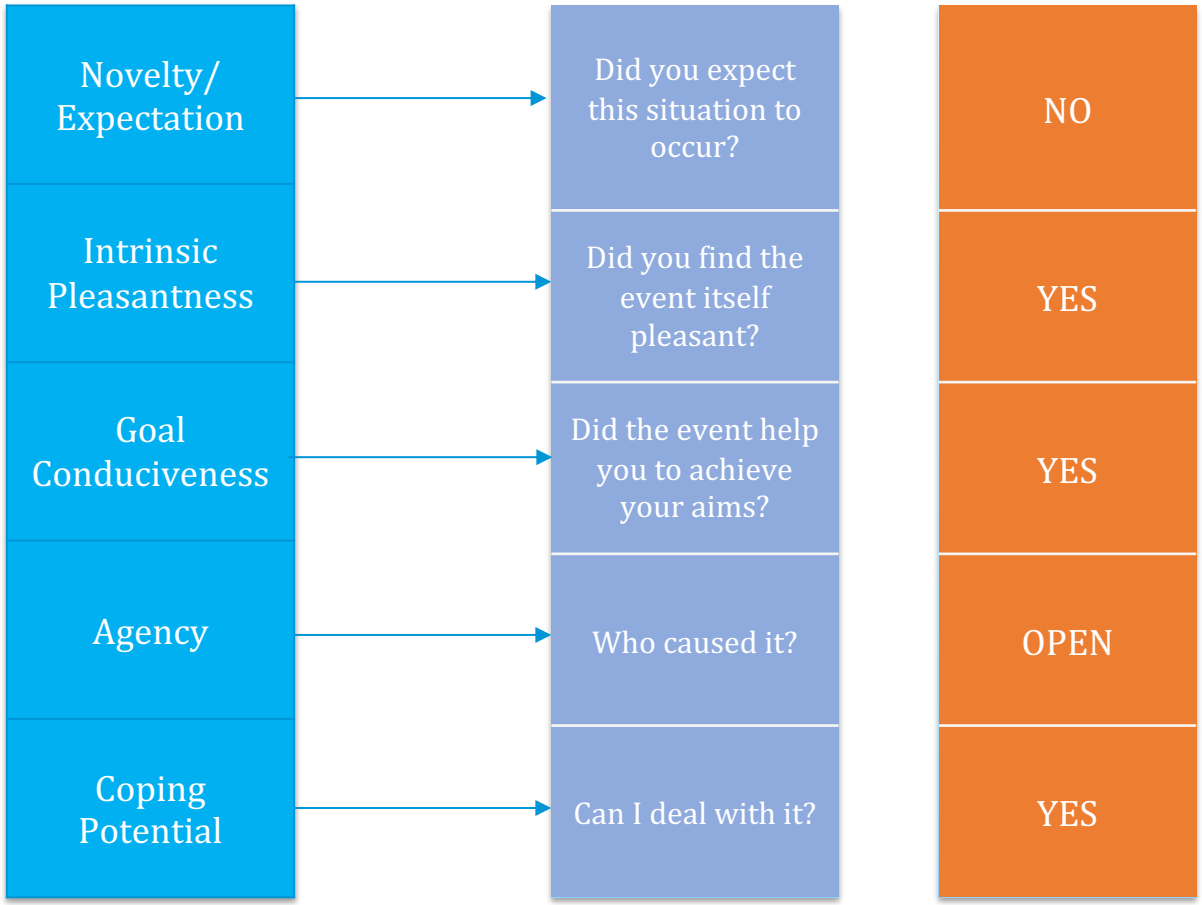
Appraisal Theories of Emotion



Appraisal Theories of Emotion



Appraisal Theories of Emotion



Surprise (positive)

Appraisal Theories of Emotion

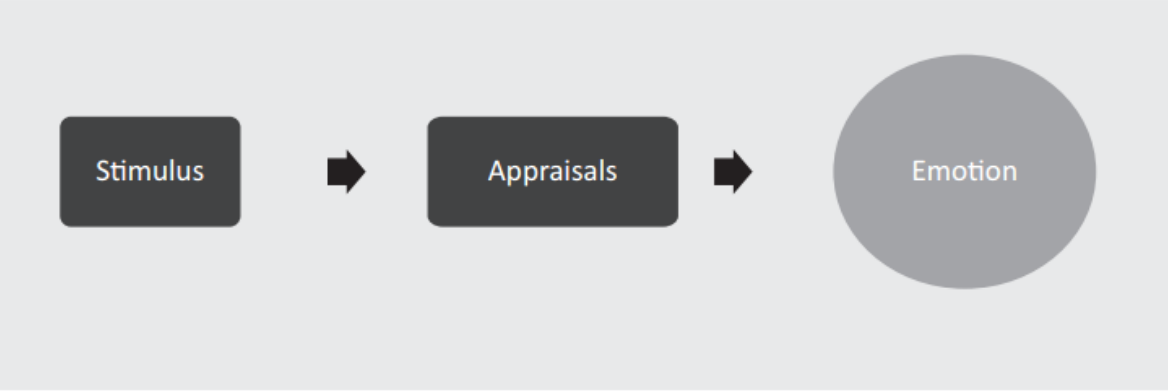
TABLE 1
Predictions on Emotion Differentiation based on Stimulus Evaluation Checks

	Joy	*Fear*	*Anger*	*Sadness*	*Disgust*	*Shame*	*Guilt*
Novelty/Expectancy							
Expectedness	Open	Low	Open	Open	Open	Open	Open
Intrinsic Pleasantness							
Unpleasantness	Low	High	Open	Open	Very high	Open	Open
Goal/Need-conduciveness							
Goal hindrance	Very low	High	High	High	Open	Open	Low
Coping Potential							
External causation	Open	External	External	Open	External	Internal	Internal
Coping ability	Medium	Very low	High	Low	Open	Open	Open
Compatibility with standards							
Immorality	Open	Open	High	Open	Open	Open	Very high
Self-consistency	Open	Open	Low	Open	Open	Very low	Very low

Note: This table is based on table 2 in Scherer (1986) and table 5 in Scherer (1988). Open = different evaluation results are compatible with the respective emotion and/or further detail (on subtype of emotion and/or subchecks) needed for predictions.

Appraisal Theories of Emotion

- **Causes:** Emotions caused by appraisals
- **Biological givens:** Novelty and valence (at least) are biological givens
- **Components:** Appraisals can affect other components, but not necessarily all together
- **Integration:** Less strong integration between components



Constructionist Theories

- William James's Theory of Emotion (1884):



- Two-Factor Theory of Emotion (1962):

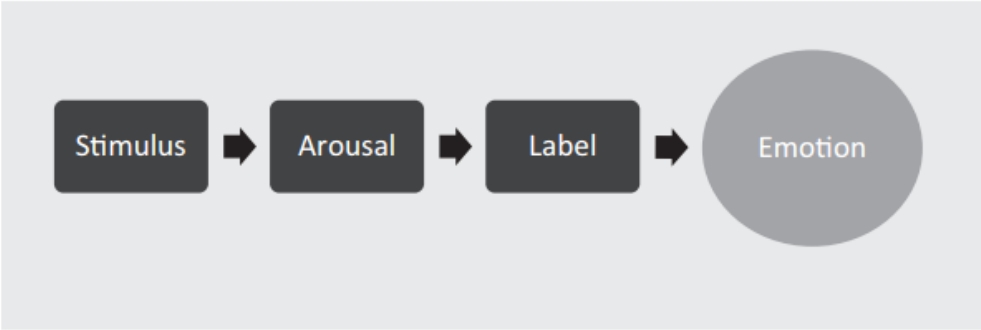
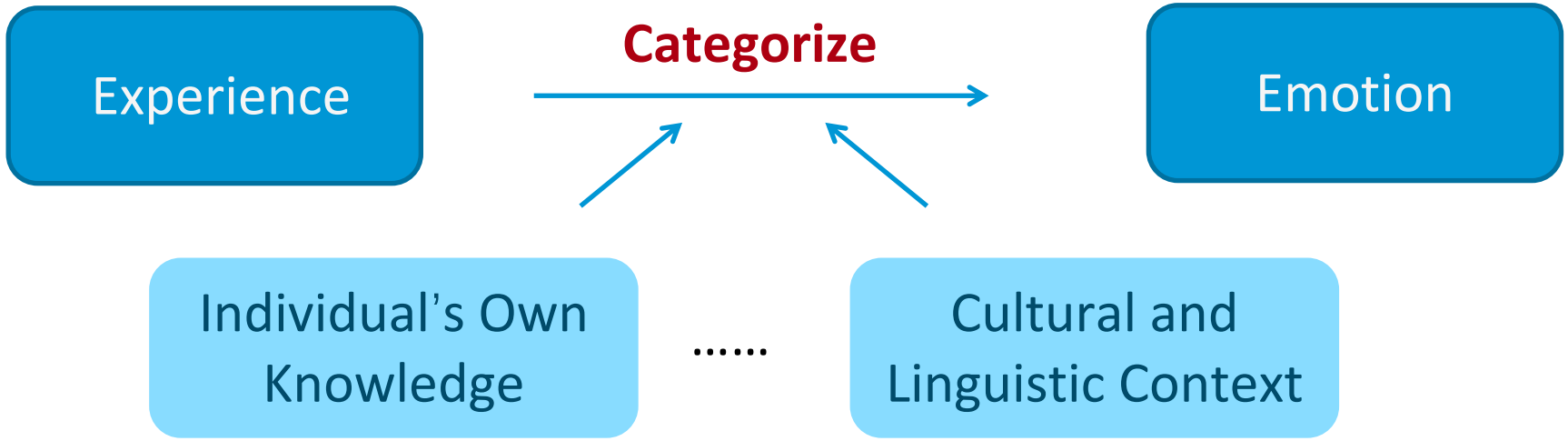


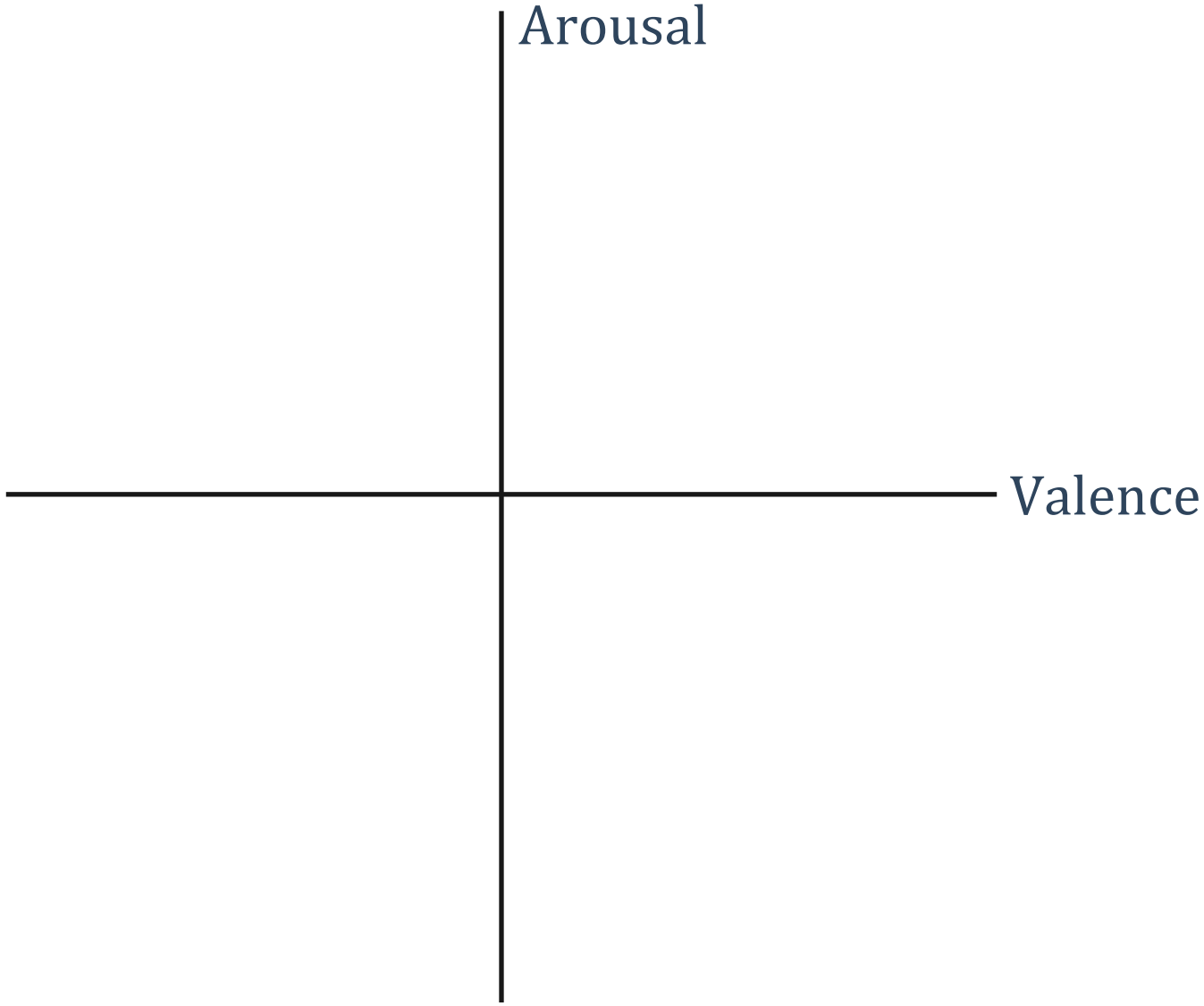
Figure 1.1 A schematic diagram of Schachter and Singer's (1962) Two-Factor Theory of emotion.

Constructionist Theories: Causes

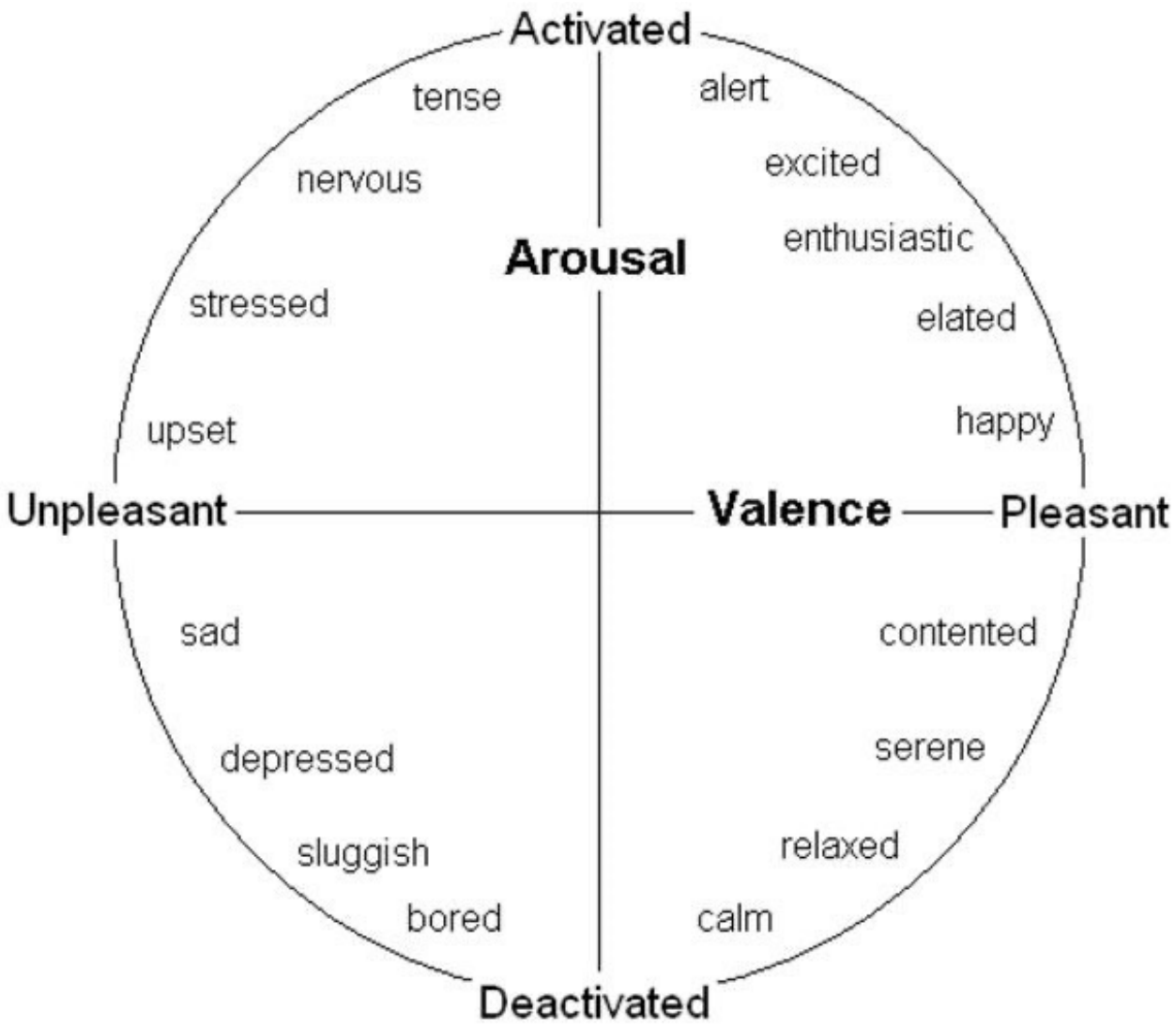
- Specific emotions are caused by applying learned categories to experience.



Constructionist Theories of Emotion

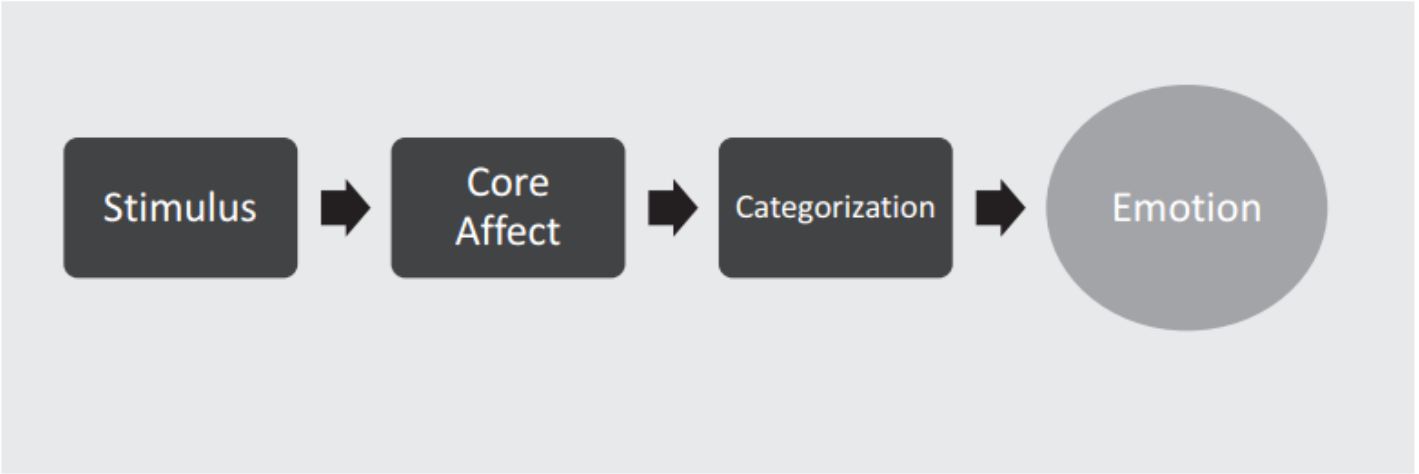


Constructionist Theories of Emotion



Constructionist Theories of Emotion

- **Causes:** Emotions caused by valence and arousal (core affect) in combination with conceptual knowledge
- **Biological givens:** Core affect is a biological given
- **Components:** Components vary across experiences; Subjective experience central
- **Integration:** Little integration between components



Constructionist Theories of Emotion

- You aren't at the mercy of your emotions



How to study emotions?

How to study emotions

- Inducing Emotions
- Measuring Emotions

How to induce emotions?

Inducing Emotions

Think about different ways to induce emotions (3 minutes)

Inducing emotions

1) Pictures
2) Movies
3) Music
4) Memories
5) Scripted social interactions

Inducing emotions - Pictures

- Validated stimulus sets
- Induces weak, short-lived emotions
- Good for multiple measures
- Works only for some emotions



Inducing emotions - Movies

- Validated stimulus sets
- Can elicit specific emotions
- Can elicit strong emotions
- Few repetitions
- Works only for some emotions



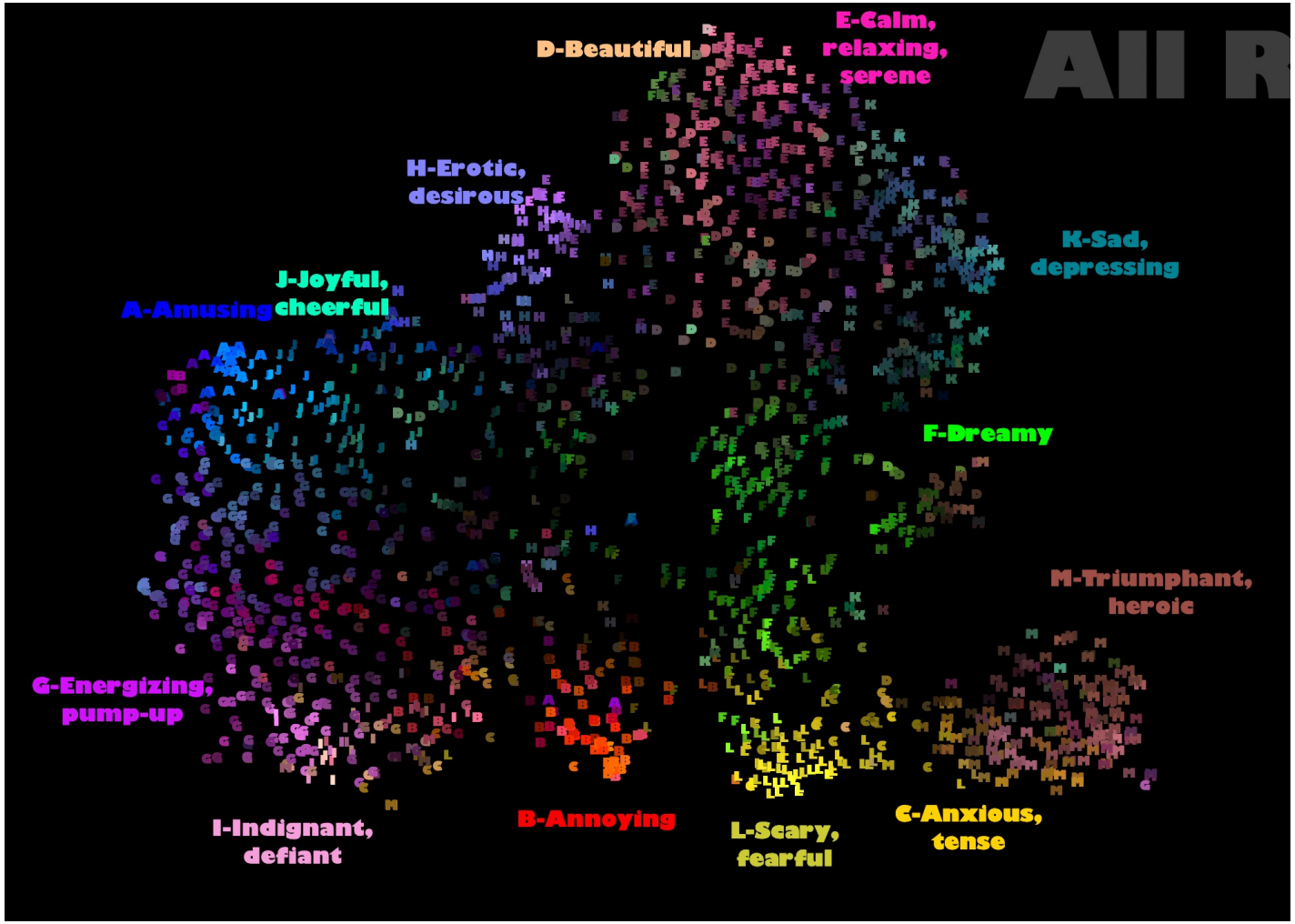
Inducing emotions - Music

- Validated stimulus sets
- Induces diffuse emotions or moods
- Works only for some emotions



Inducing emotions - Music

Music can elicit **13** emotions across USA and China.



Inducing emotions - Memories

- Can elicit intense emotions
- Can elicit specific emotions
- Few repetitions
- Works for most emotions



Inducing emotions - Scripted social interactions

- Can elicit intense emotions
- Can elicit specific emotions
- Typically only one trial per participant
- Works only for some emotions

Inducing emotions - Naturally occurring emotions





How to measure emotions?

Measuring Emotions

Think about different ways to measure emotions (3 minutes)

Measuring emotions

1) Self-report (questionnaires)
2) Biological measures
3) Nonverbal expressions

Measuring emotions - Self-report (questionnaires)

How angry do you feel?

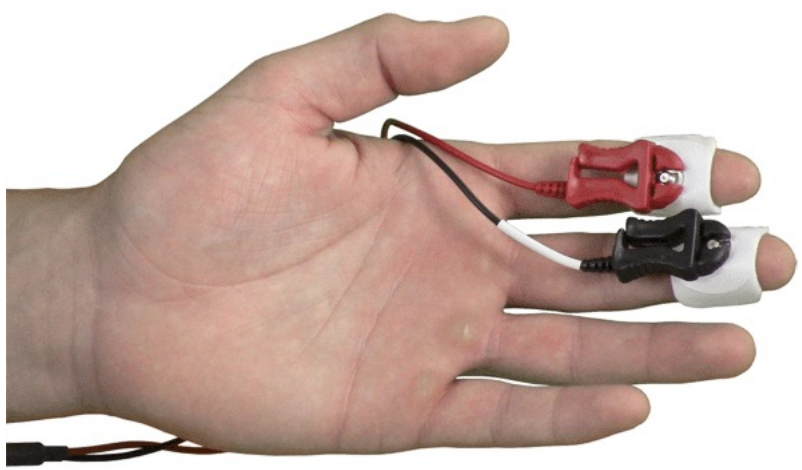
1……2……3……4……5……6……7……8……9

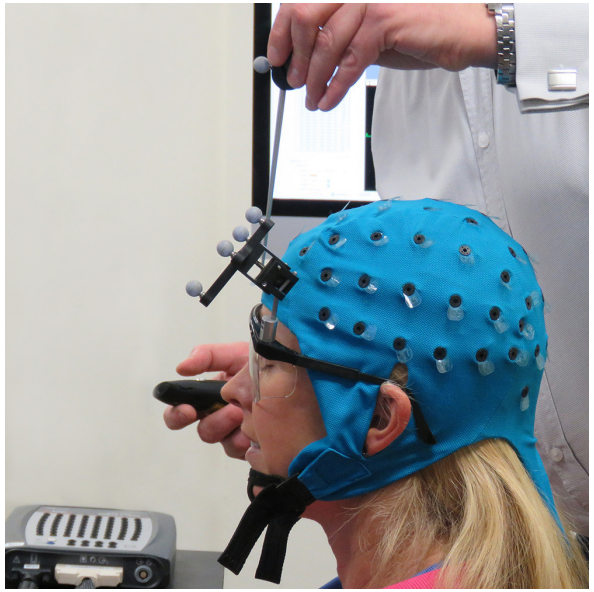
Not at all angry

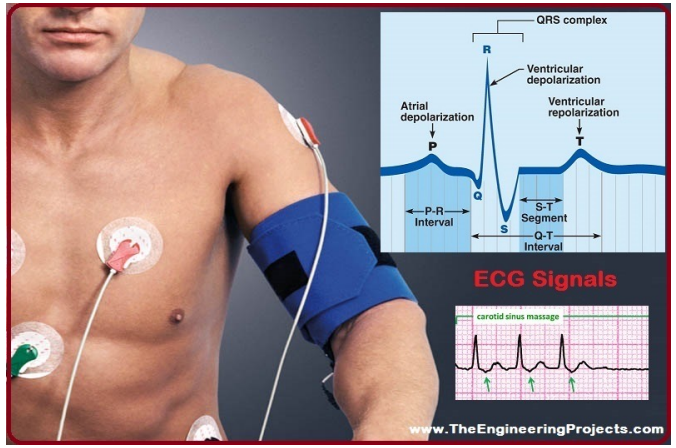
Extremely angry

Measuring emotions - Biological measures

1) Skin Conductance (EDA/GSR)
2) Brain Activity (EEG, fMRI)
3) Heart Rate (ECG)







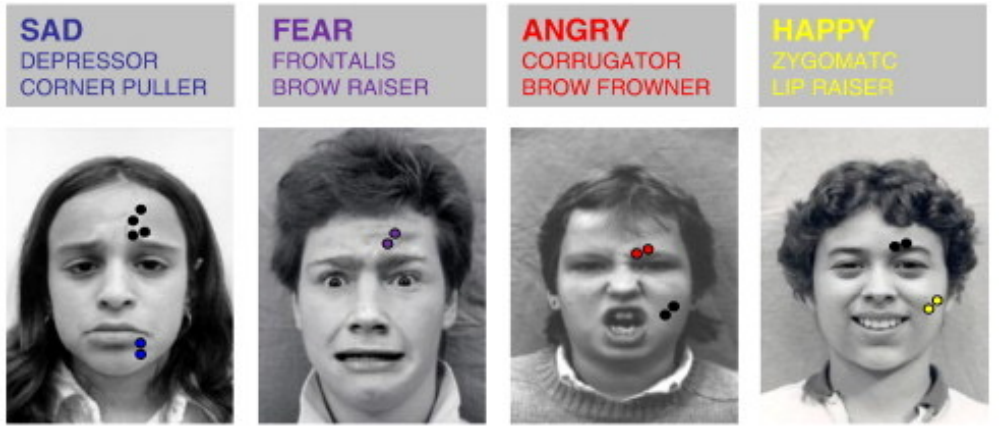
Measuring emotions - Nonverbal expressions

1) Facial Expressions
   a) Facial EMG
   b) Automatic FACS-based Software: FaceReader, iMotion,
   c) Manual FACS Coding
2) Pupil Dilation (eye tracking)
3) Gesture
4) Touch
5) Voice

Measuring emotions - Nonverbal expressions

1) Facial Expressions
   a) Facial EMG
   b) Automatic FACS-based Software: FaceReader, iMotion,
   c) Manual FACS Coding





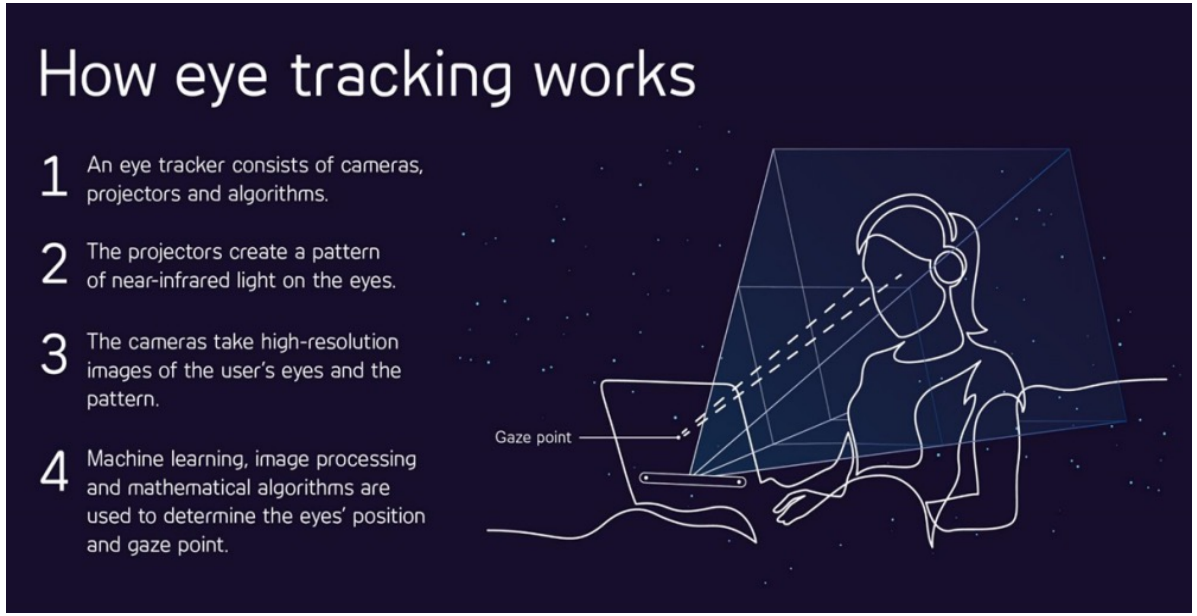
	depressor	frontalis	corrugator	zygomaticus
SAD	+	+	+	
FEAR		+		
ANGRY			+	-
HAPPY			-	+

Measuring emotions - Nonverbal expressions

2) Pupil Dilation (eye tracking)

演示





Measuring emotions - Nonverbal expressions

3) Gesture



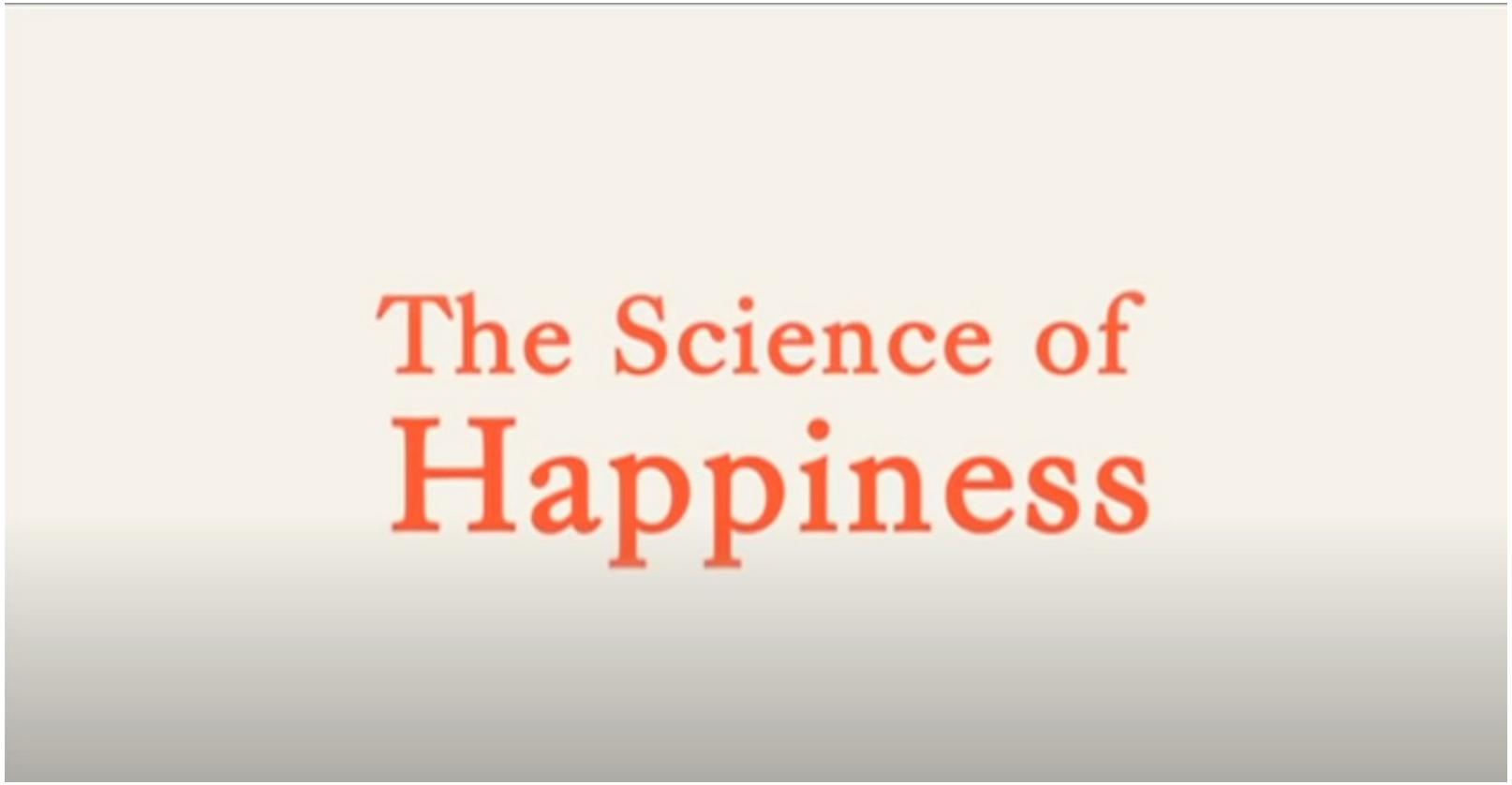
The Body Action and Posture coding system (BAP)





Measuring emotions - Nonverbal expressions

4) Touch



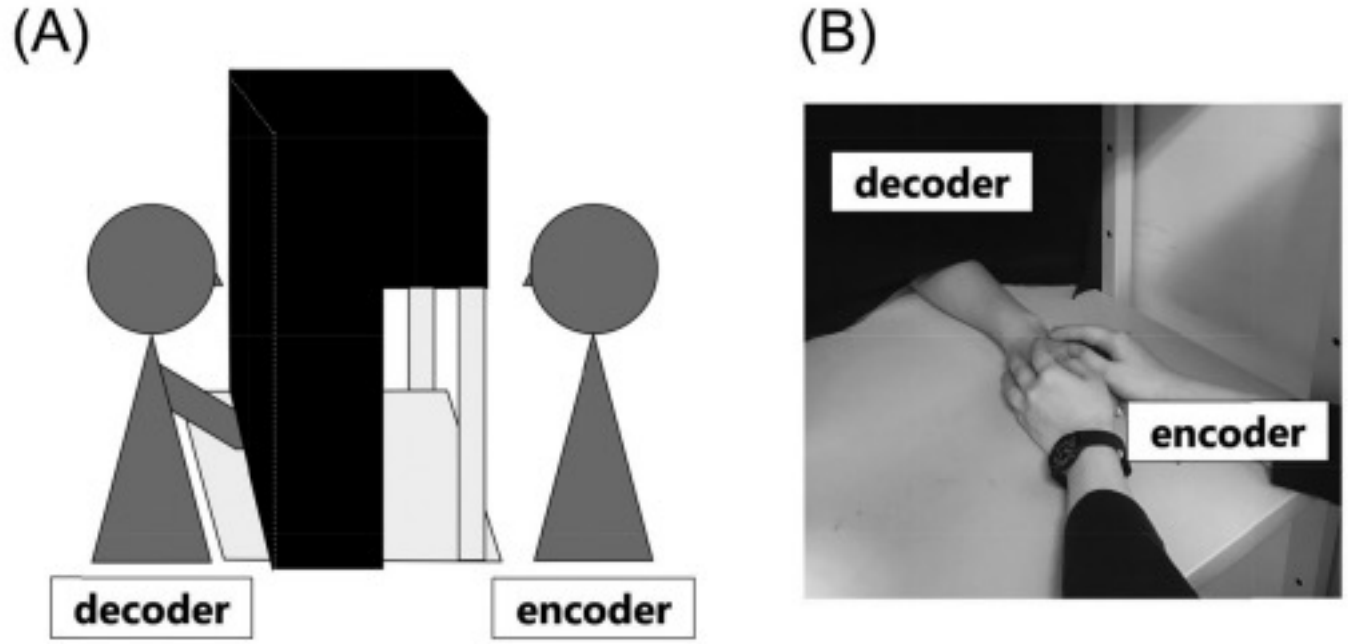
Measuring emotions - Nonverbal expressions

4) Touch



Measuring emotions - Nonverbal expressions

4) Touch



Measuring emotions - Nonverbal expressions

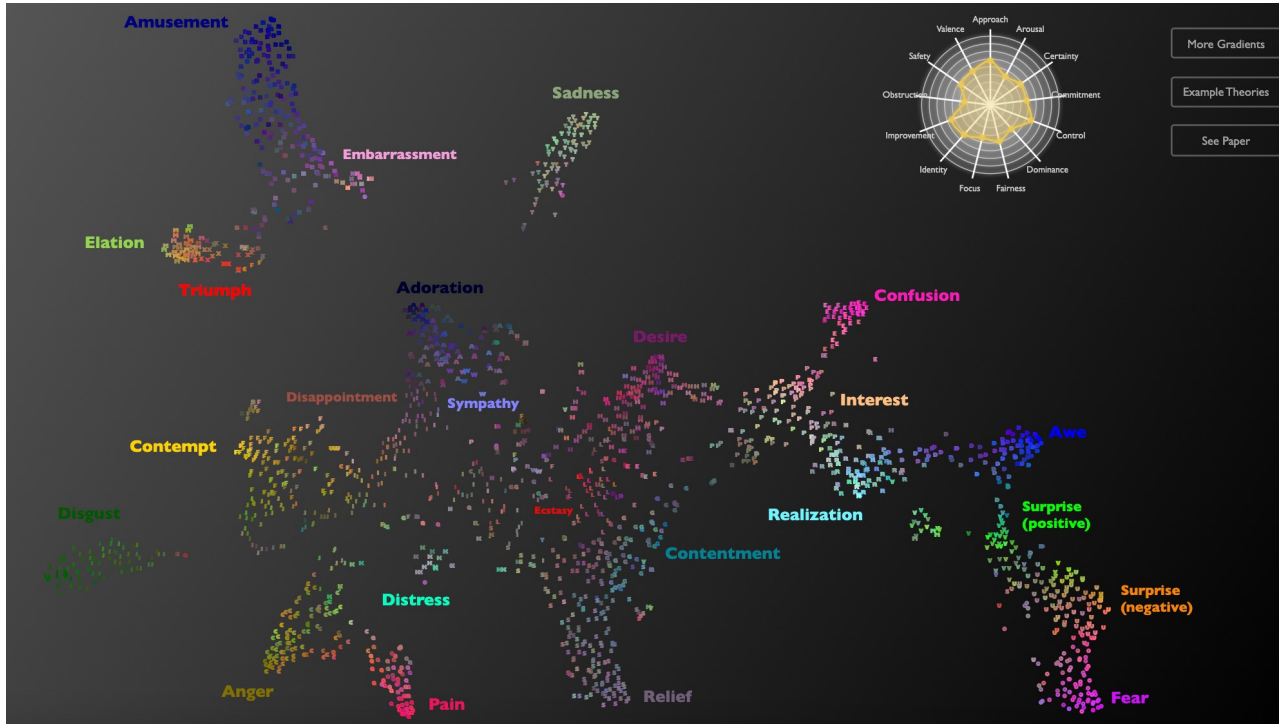
4) Touch

Sympathy, Gratitude, Love, Anger, Fear, Disgust

Dacher Keltner

Measuring emotions - Nonverbal expressions

5) Voice
   a) Vocalization（叹词）？
   b) Speech Prosody（韵律）？



Chinese Emotional Speech Prosody and Vocal Bursts

Participants: **62** Chinese participants (*M* = 21.48, *SD* = 2.42; 31 females)

vocal bursts (32)

Adoration	喜爱	Interest	兴趣
Amusement	好笑	moved	感动
Anger	气愤	Pain (physical)	痛苦
Awe	令人惊叹	praise	赞扬
Confusion	困惑	pride	自豪/骄傲
Contempt	蔑视	Realization	恍然大悟
Contentment	满足	Relief	解脱
Desire	欲望	Sadness	悲伤
Desire (for food)	性欲	*Sajiao*	撒娇
Desire (for sex)	食欲	shame/guilt	愧疚/内疚
Disgust	厌恶	snicker/grin	坏笑
Disgust (physical)	恶心	snigger	偷笑
Distress	苦恼	Surprise	惊讶
Embarrassment	尴尬	Surprise (positive)	惊喜
Excitement	兴奋	Sympathy	同情
Fear	恐惧	Triumph	胜利

speech prosody (25)

Adoration	喜爱	Interest	兴趣
Amusement	好笑	moved	感动
Anger	气愤	Neutral	中性
Awe	令人惊叹	praise	赞扬
Confusion	困惑	pride	自豪/骄傲
Contempt	蔑视	Sadness	悲伤
Desire	欲望	*Sajiao*	撒娇
Disgust	厌恶	shame/guilt	愧疚/内疚
Distress	苦恼	snicker/grin	坏笑
Excitement	兴奋	Surprise	惊讶
Fear	恐惧	Surprise (positive)	惊喜
Gratitude	感谢	Triumph	胜利
Happiness	开心		

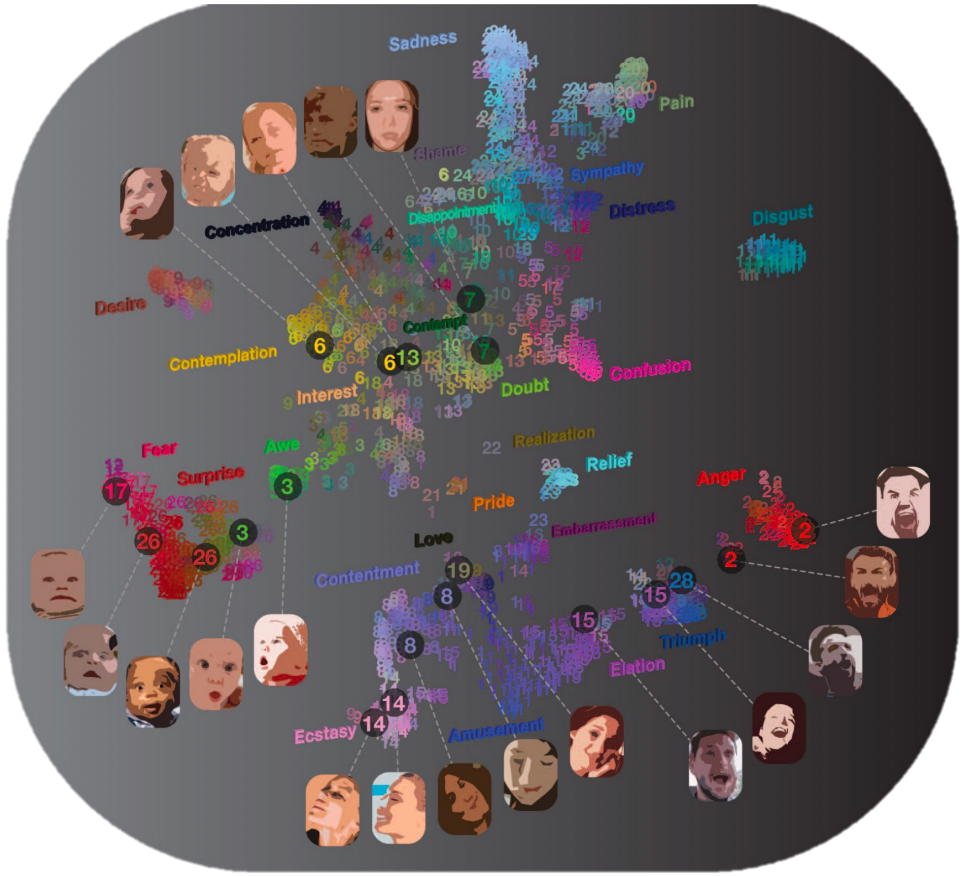


Facial Expressions of Emotion

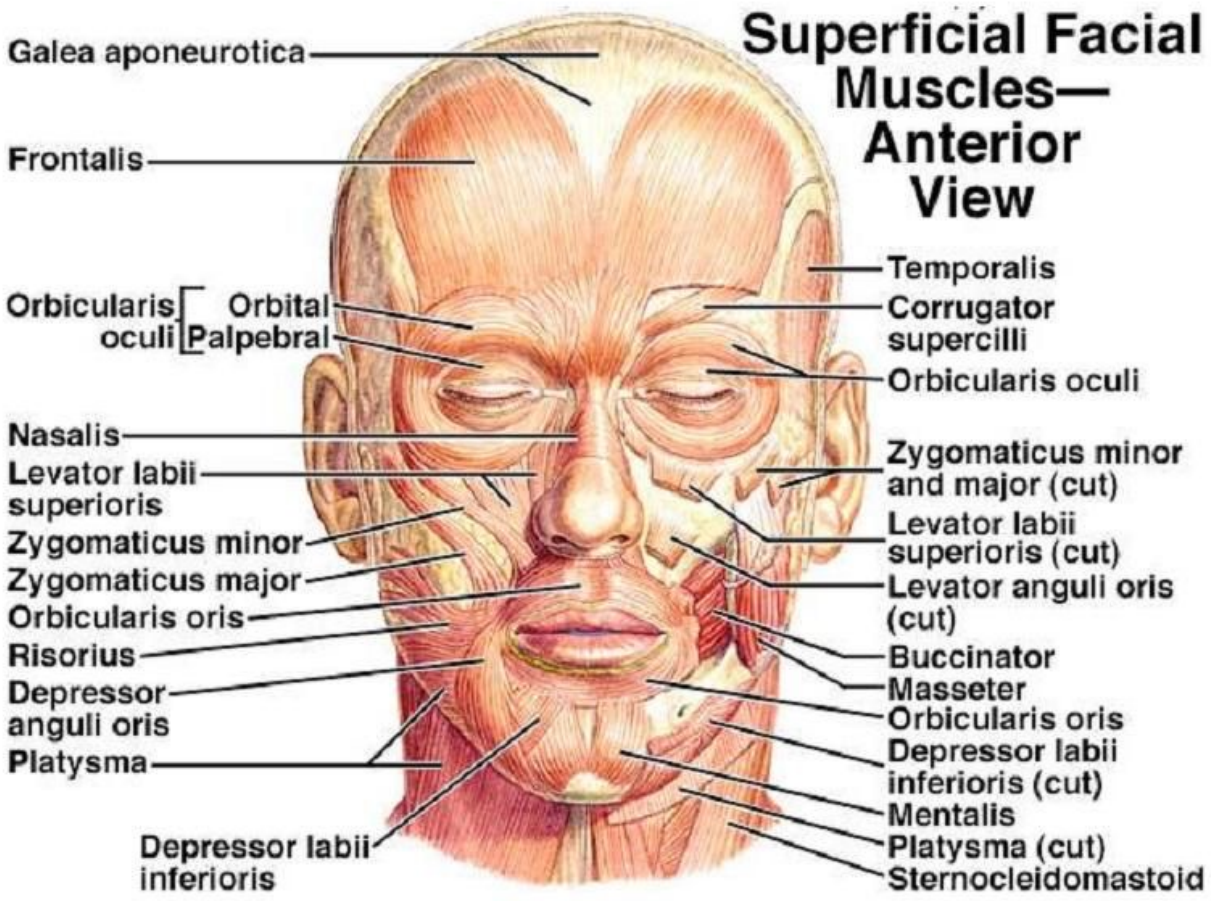
Facial expressions

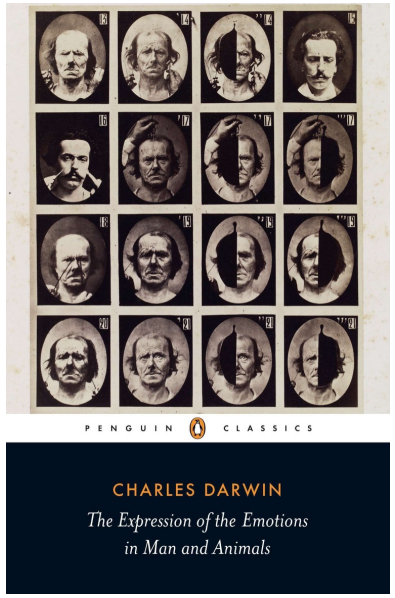
How many facial expressions of emotion are there?

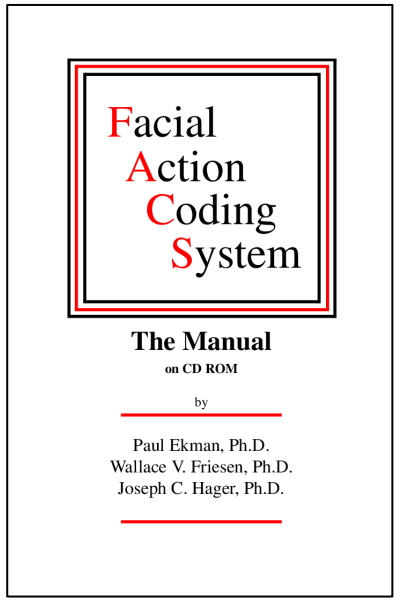
A study on the face and including nearby hand and body movements showed that there are 28 facial-bodily expressions of emotion in the United States (Cowen & Keltner, 2019).



Facial muscles







1872
Charles Darwin

1978
Paul Ekman

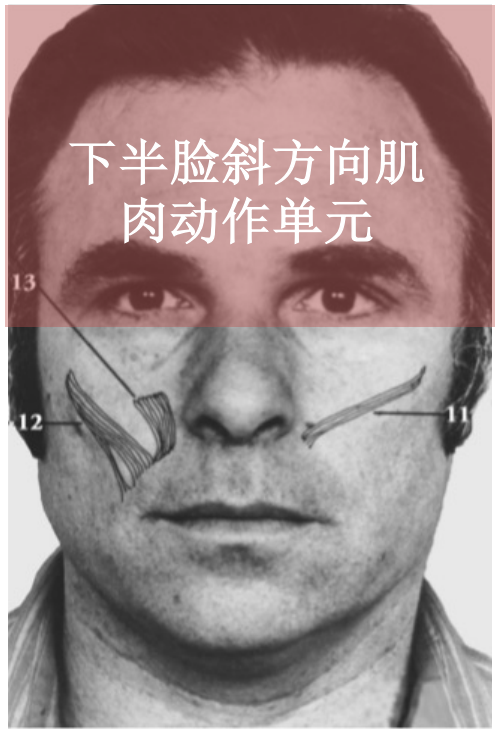
2023

面部动作编码系统
(Facial Action Coding System)









44 个面部动作单元

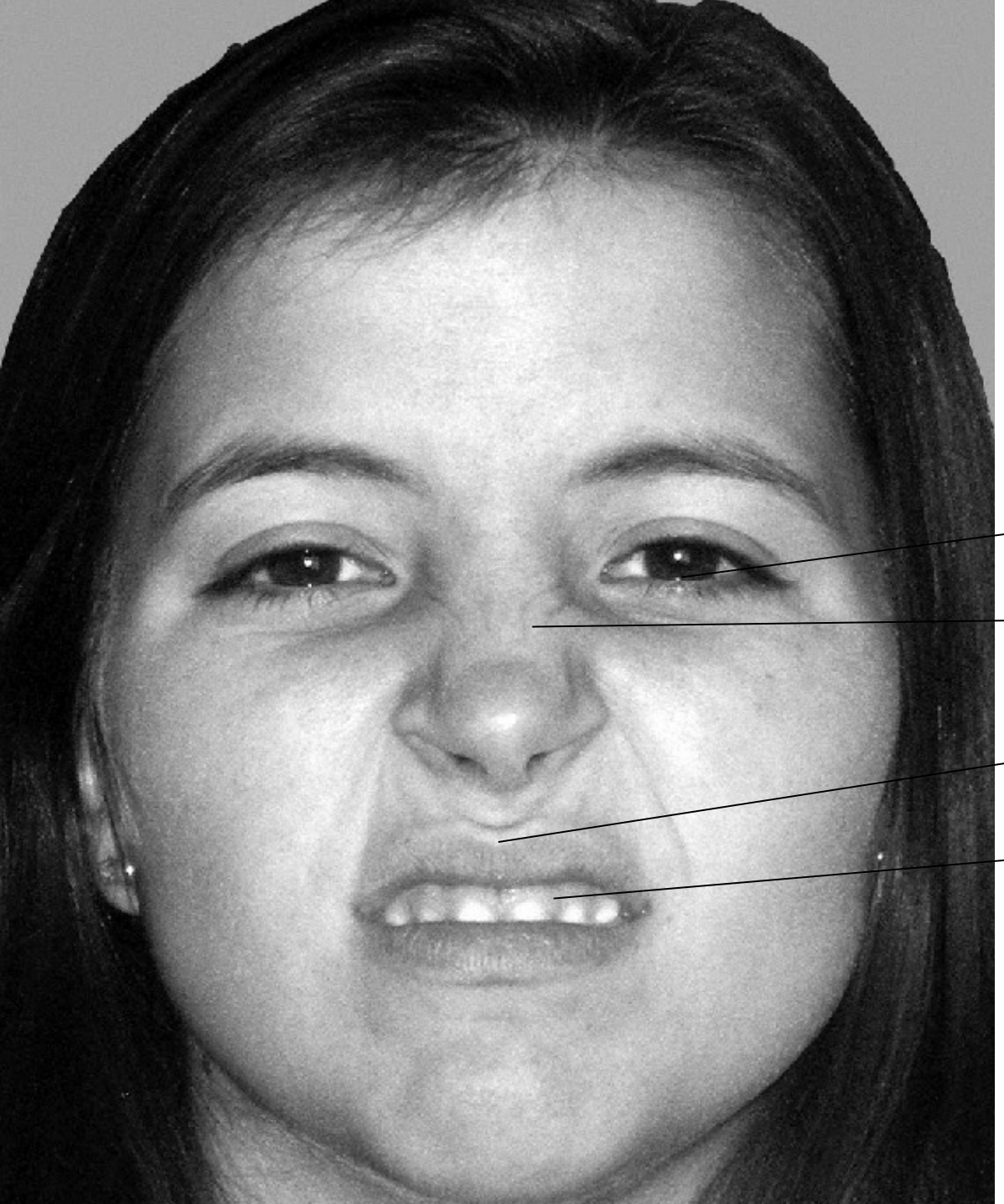
Facial Action Coding System (FACS)

Upper Face Action Units					
AU 1	AU 2	AU 4	AU 5	AU 6	AU 7
					
Inner Brow Raiser	Outer Brow Raiser	Brow Lowerer	Upper Lid Raiser	Cheek Raiser	Lid Tightener
*AU 41	*AU 42	*AU 43	AU 44	AU 45	AU 46
					
Lid Droop	Slit	Eyes Closed	Squint	Blink	Wink
Lower Face Action Units					
AU 9	AU 10	AU 11	AU 12	AU 13	AU 14
					
Nose Wrinkler	Upper Lip Raiser	Nasolabial Deepener	Lip Corner Puller	Cheek Puffer	Dimpler
AU 15	AU 16	AU 17	AU 18	AU 20	AU 22
					
Lip Corner Depressor	Lower Lip Depressor	Chin Raiser	Lip Puckerer	Lip Stretcher	Lip Funneler
AU 23	AU 24	*AU 25	*AU 26	*AU 27	AU 28
					
Lip Tightener	Lip Pressor	Lips Part	Jaw Drop	Mouth Stretch	Lip Suck



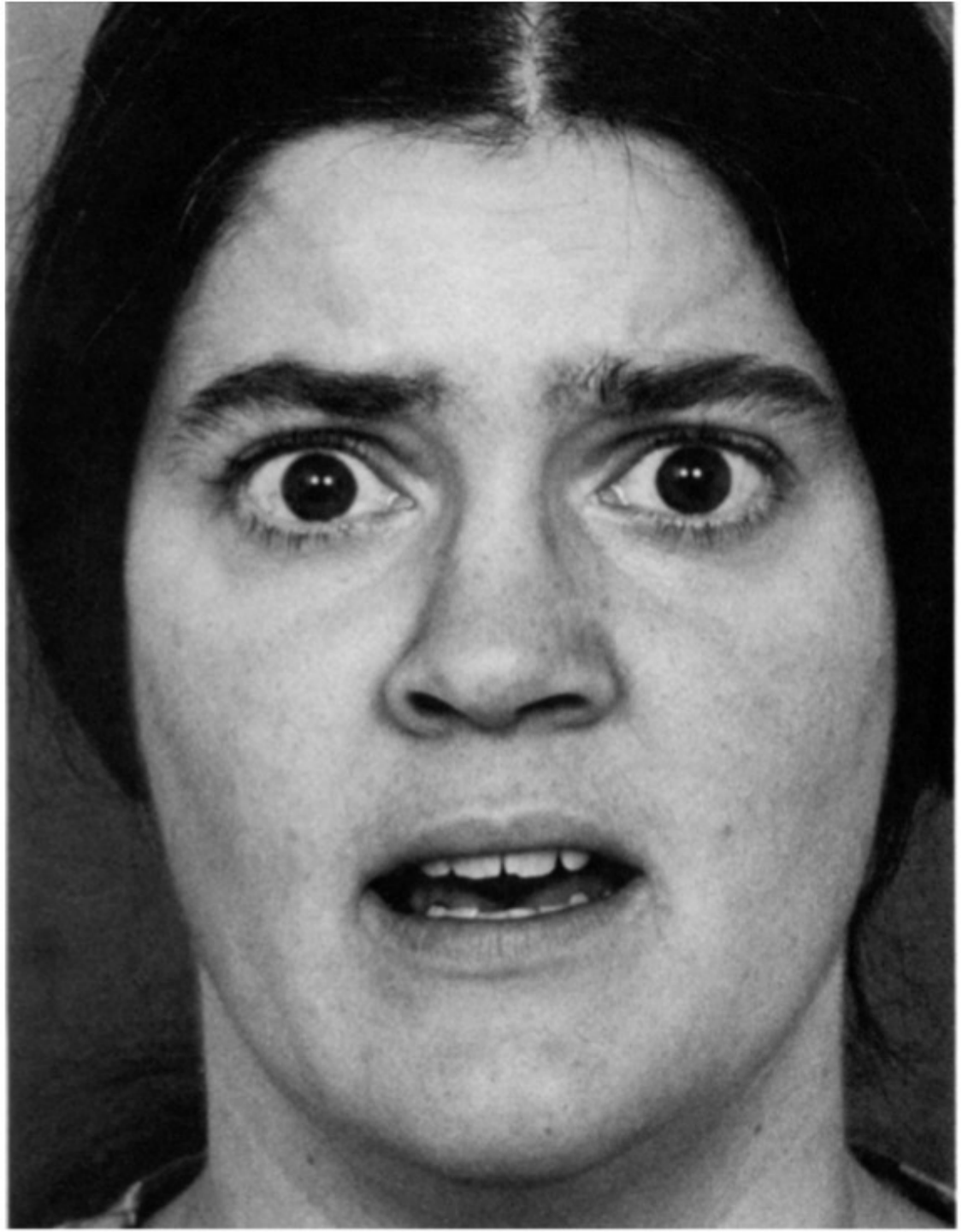
肌肉动作单元
(Action Unit)

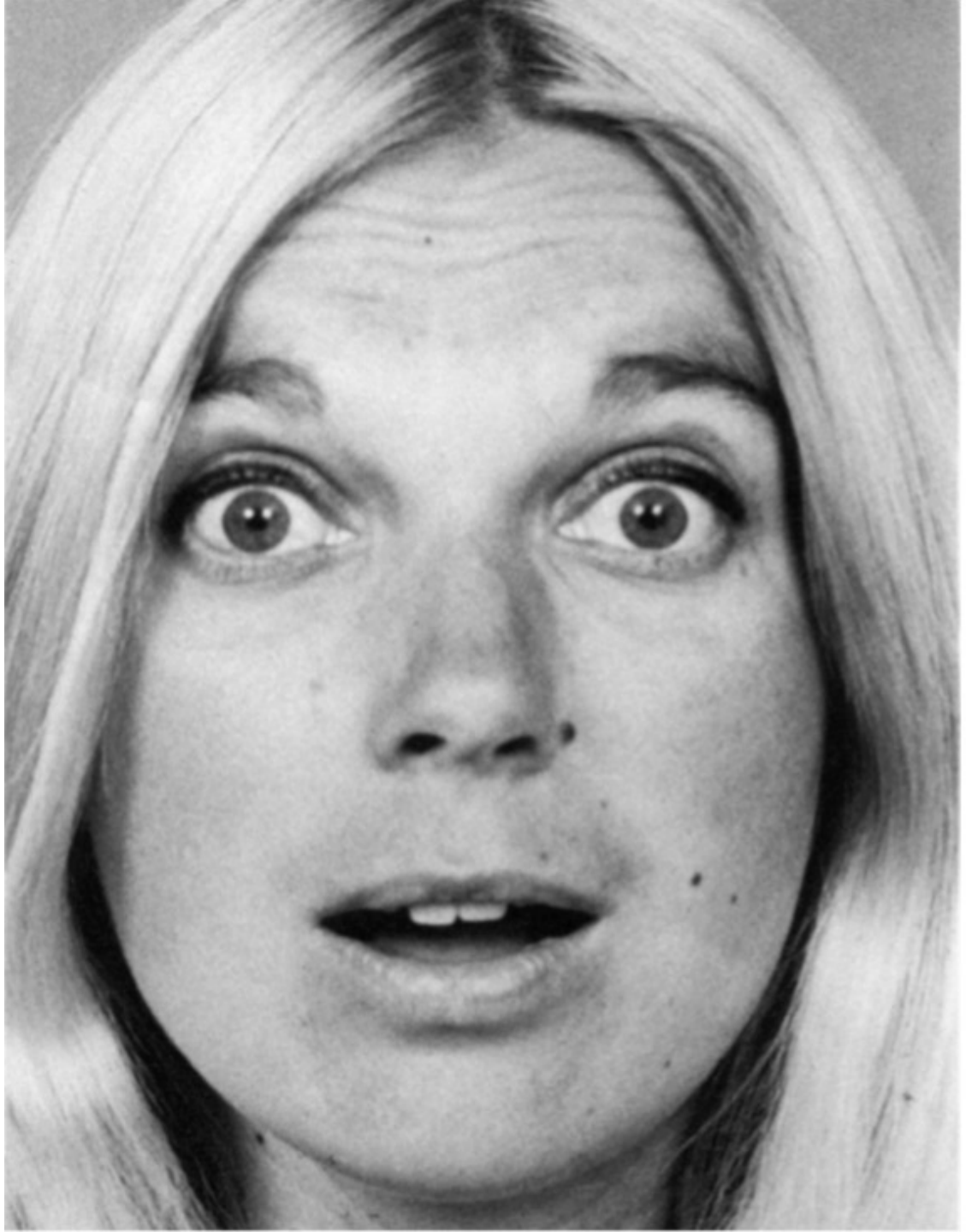
- AU4：皱眉
- AU5：上眼睑上抬
- AU24：上下唇按压
- AU17：下巴上抬



肌肉动作单元
(Action Unit)

- AU7：内眼眶缩紧
- AU9：皱鼻
- AU10：上嘴唇上抬
- AU25：上下唇分开

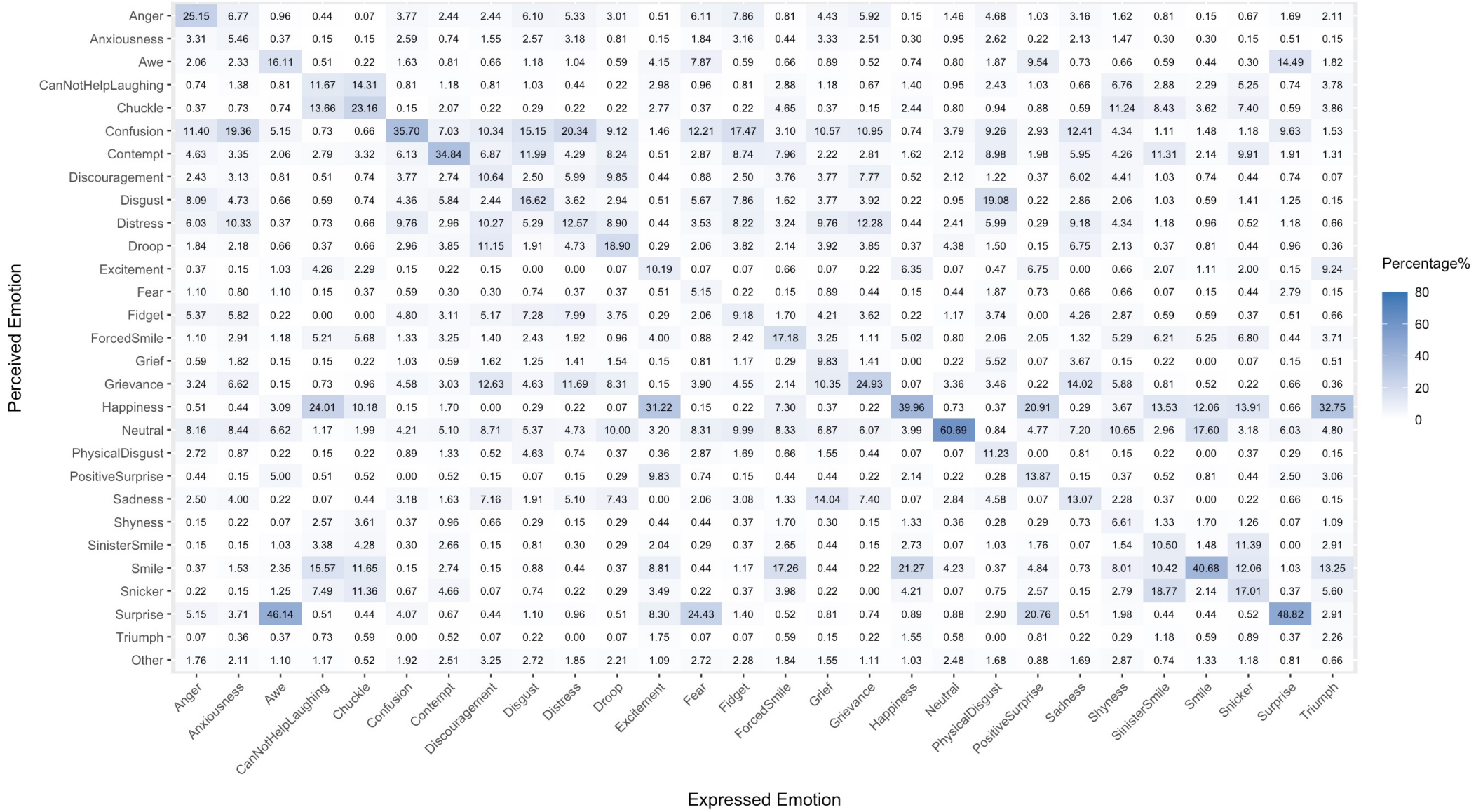




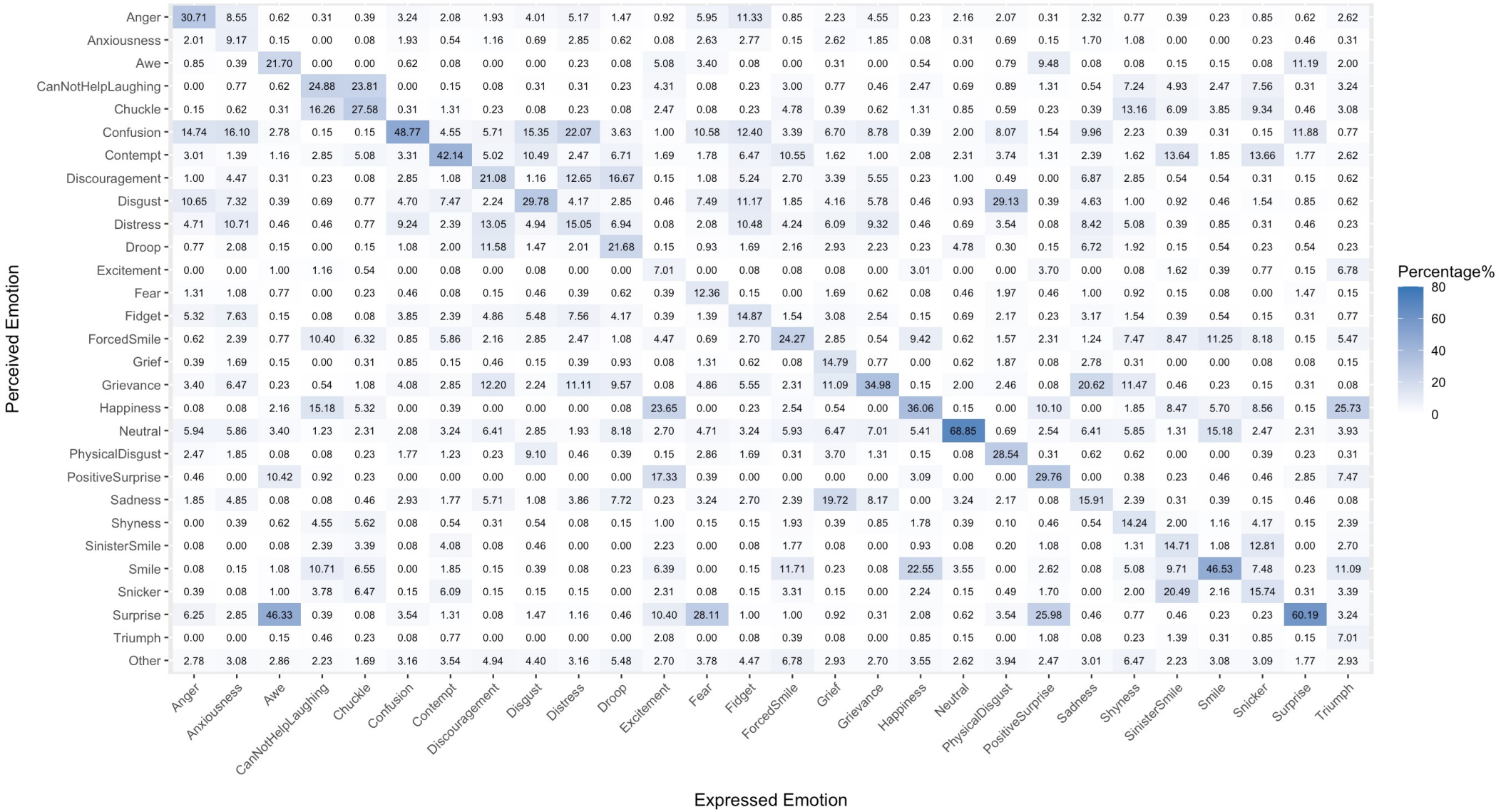
Collecting Chinese Emotional Faces



Confusion matrix for static facial expression

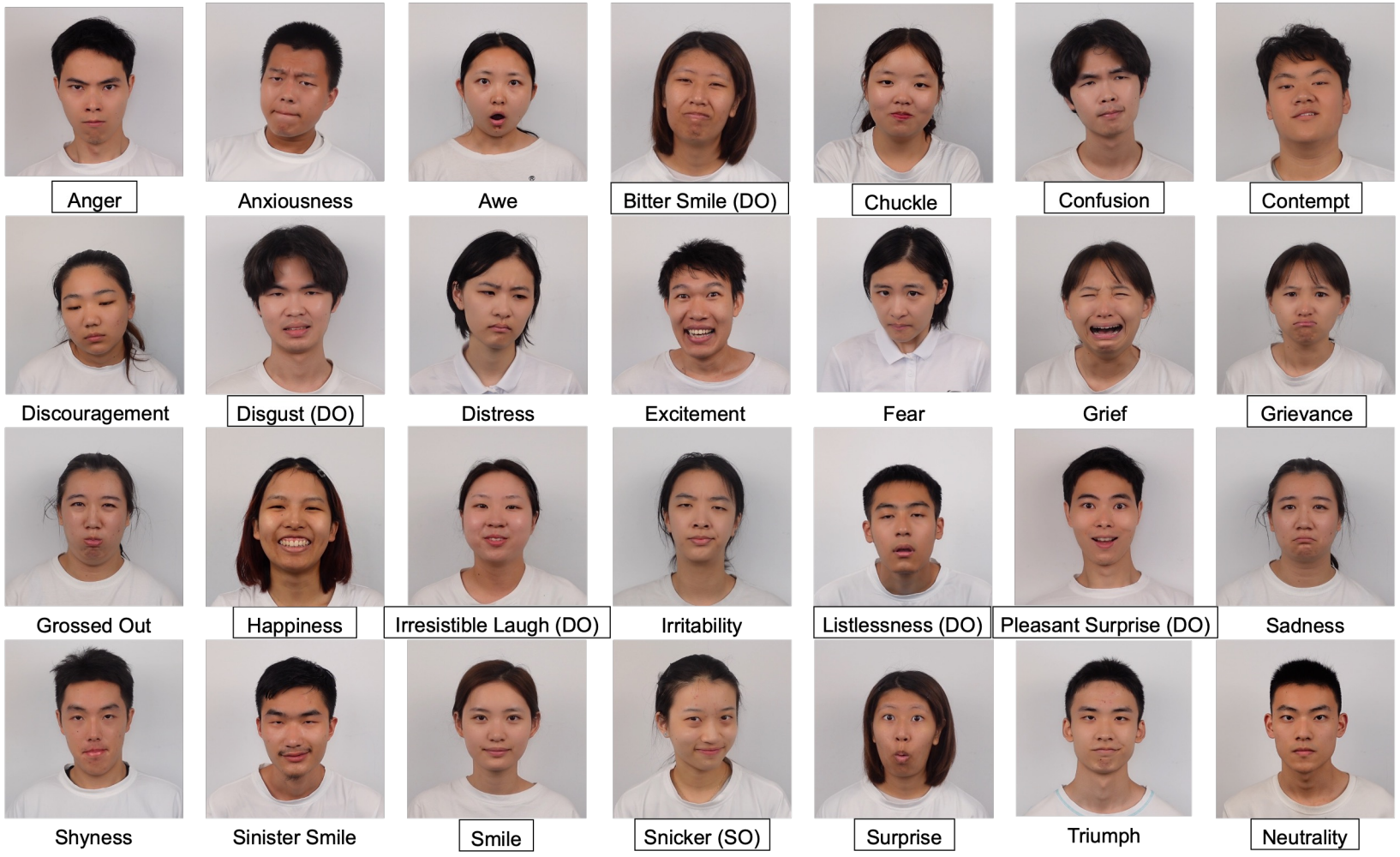


Confusion matrix for dynamic facial expression



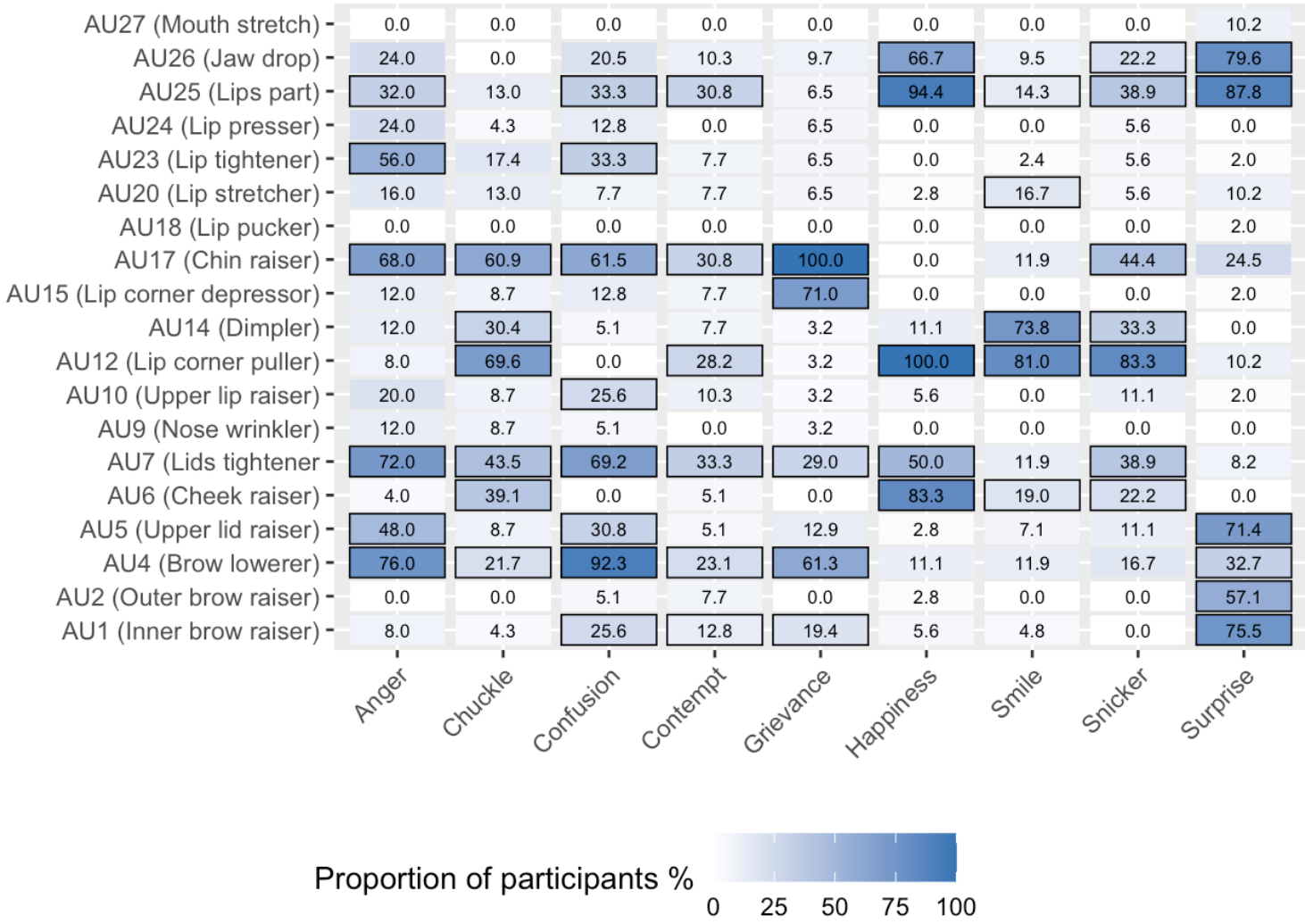
5.1 Emotions that could be conveyed through dynamic and static facial expressions

C. Examples of Dynamic and Static Facial Expression with High Recognition Rates



Participants could identify 10 emotions from static facial expressions, but 14 emotions from dynamic facial expressions.

B. Static Expressions




Child 14, happy
Child 13, angry
Child 9, neutral
Child 7, sad
Child 0, neutral
Child 10, neutra
Child 4, neutral
Child 12, neutral
Child 1, neutral
Child 11, neutra


Stu 11, happy
Stu 10, happy
Stu 1, happy
Stu 9, ne
Stu 0, happy
Stu 4, happy



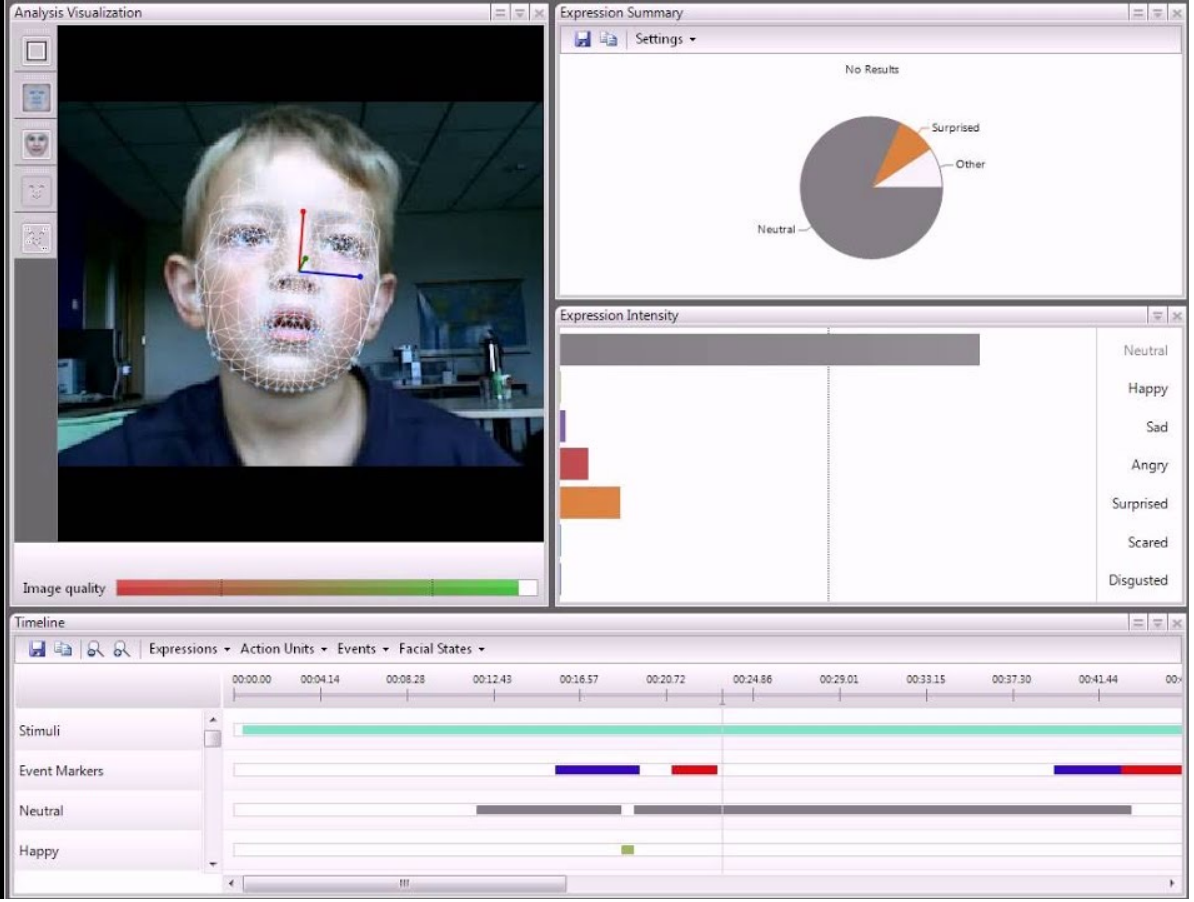

DISTRACTION 100
DROWSY 80
YAWN 0
EYE CLOSURE 100
JOY 0




pepper

Do faces always tell the truth about feelings ?



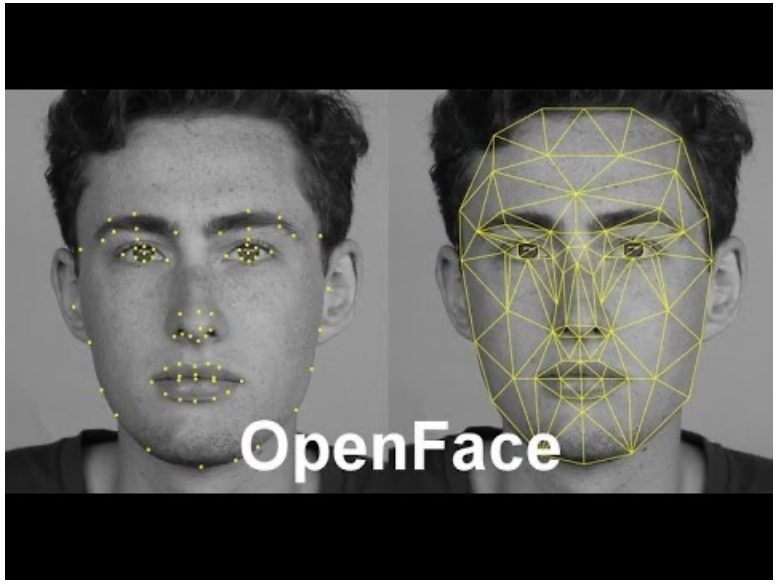


EMOTIENT®

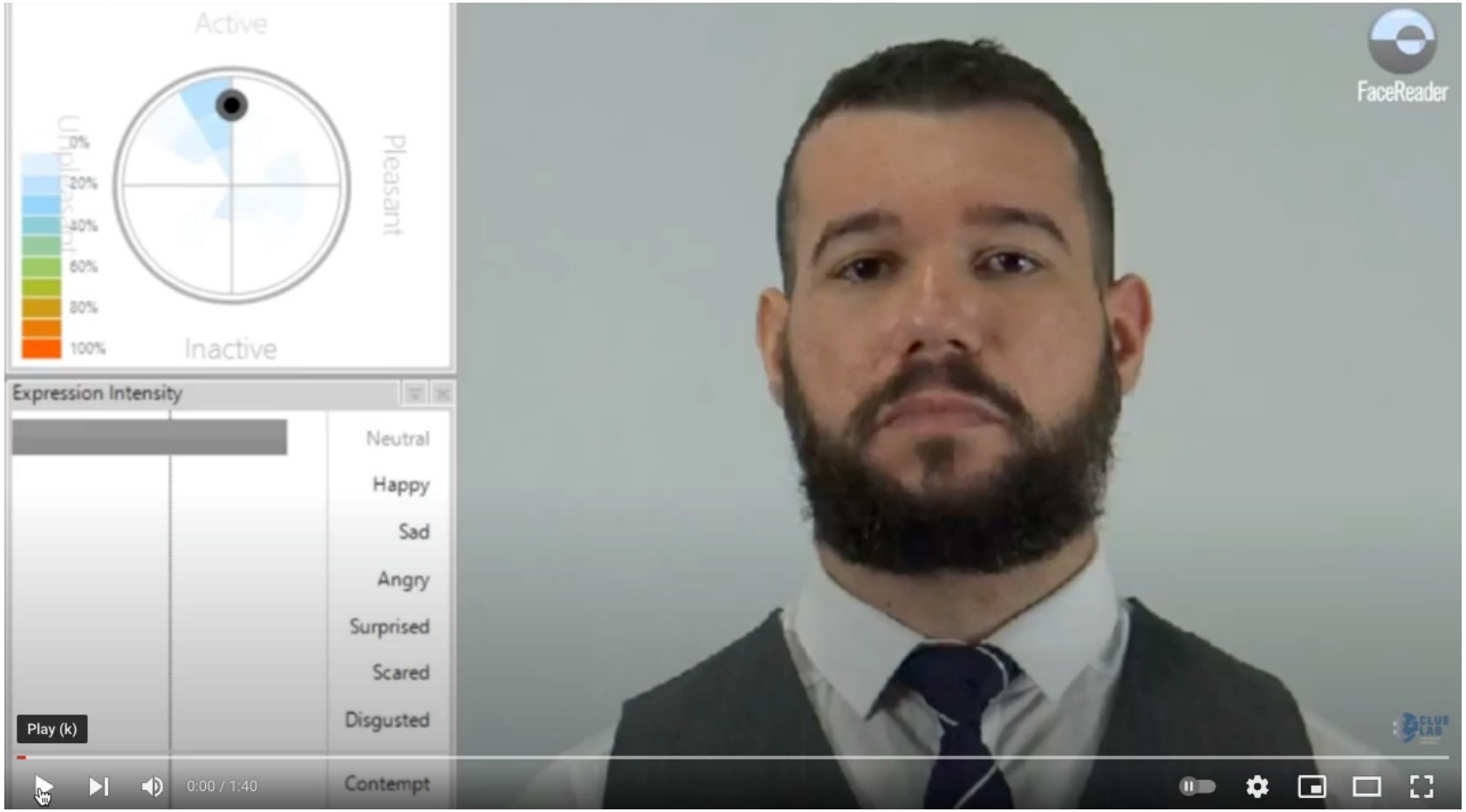
:) Affectiva

Noldus

Information Technology



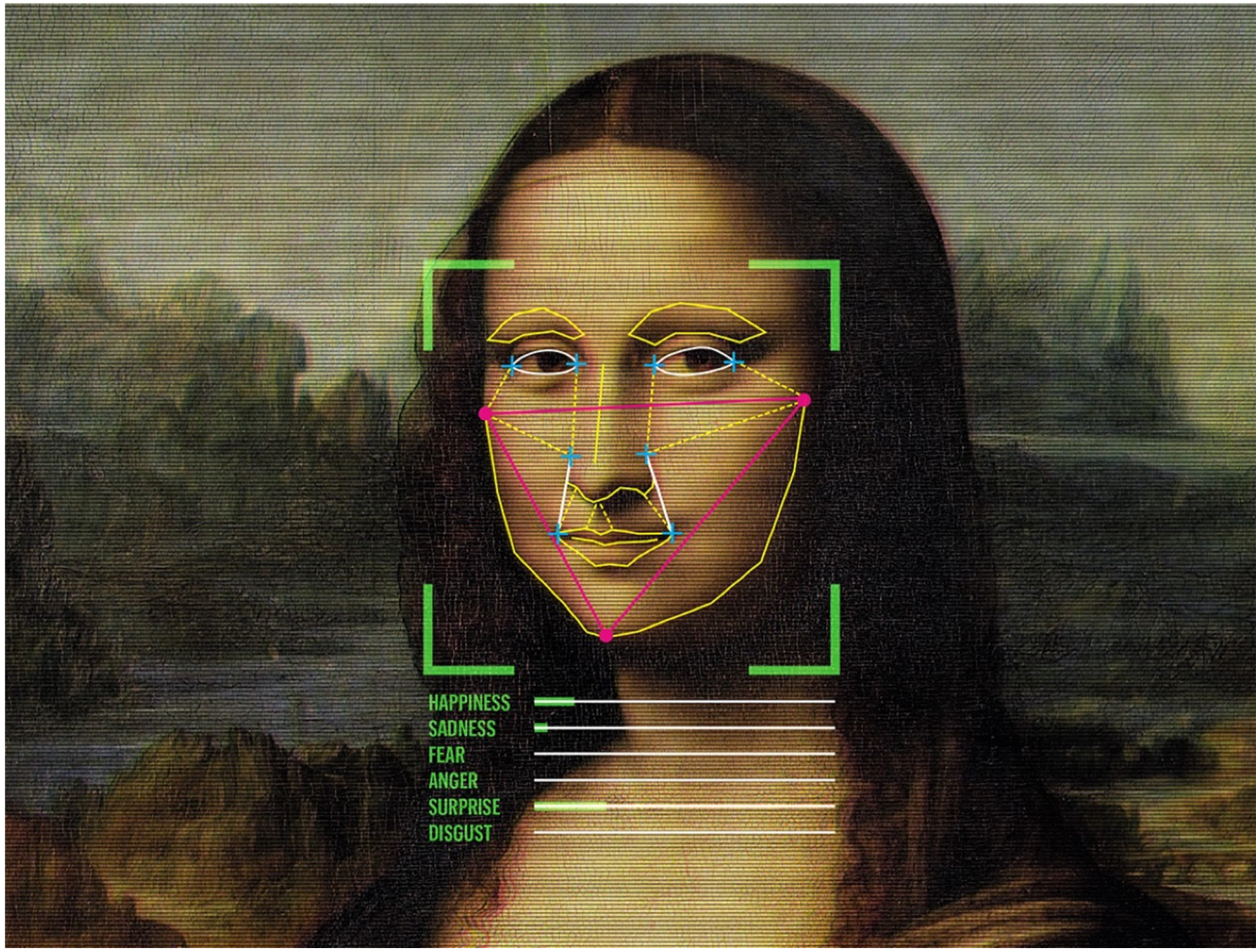
Decoding Facial Expressions



Automatic FACS Coding

问题：

- 识别情绪种类和面部动作有限；比如FaceReader目前只能判定7种表情，20个面部动作。
- 对自然情境下的自发表情识别信度和效度较低。

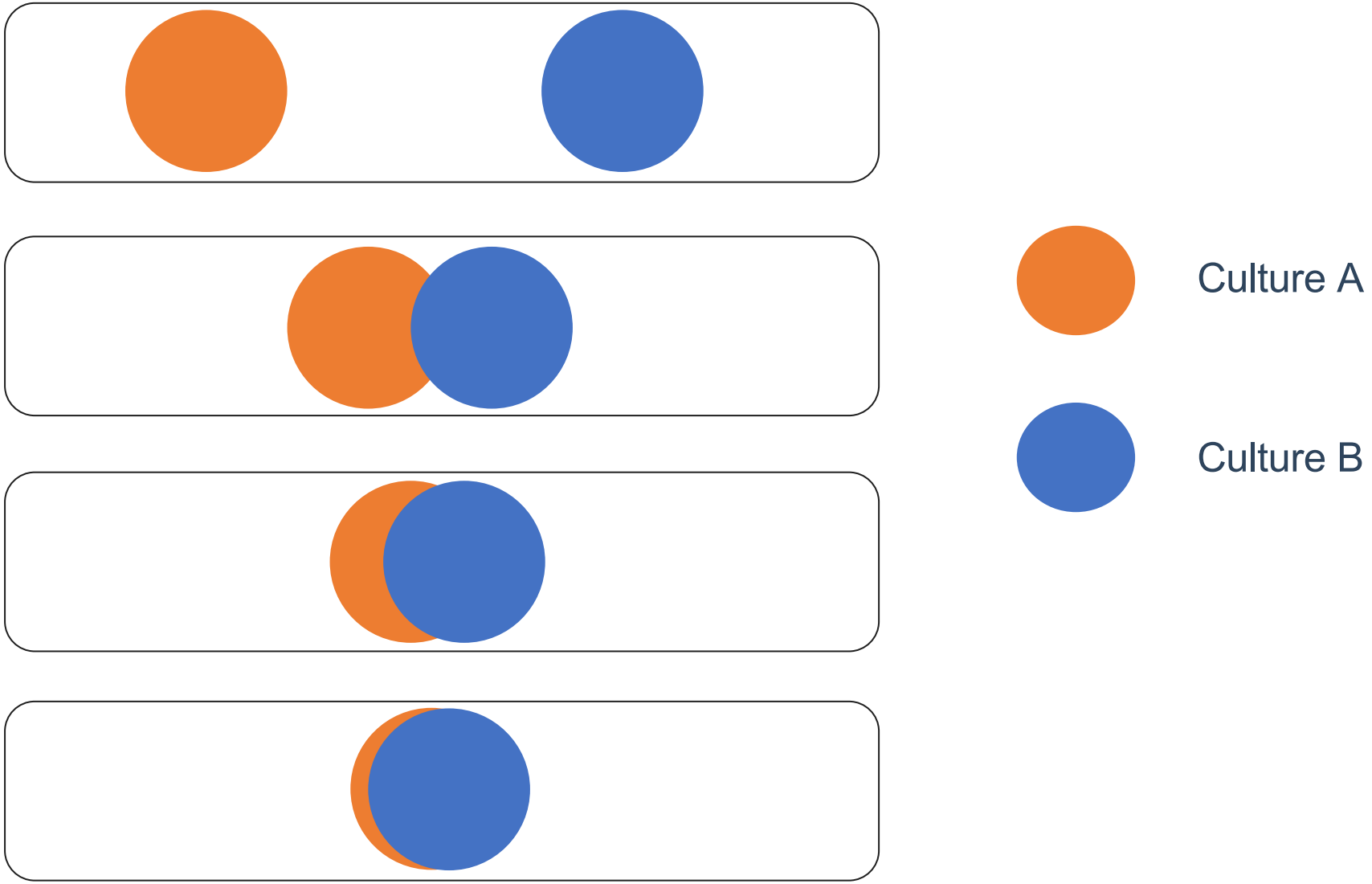


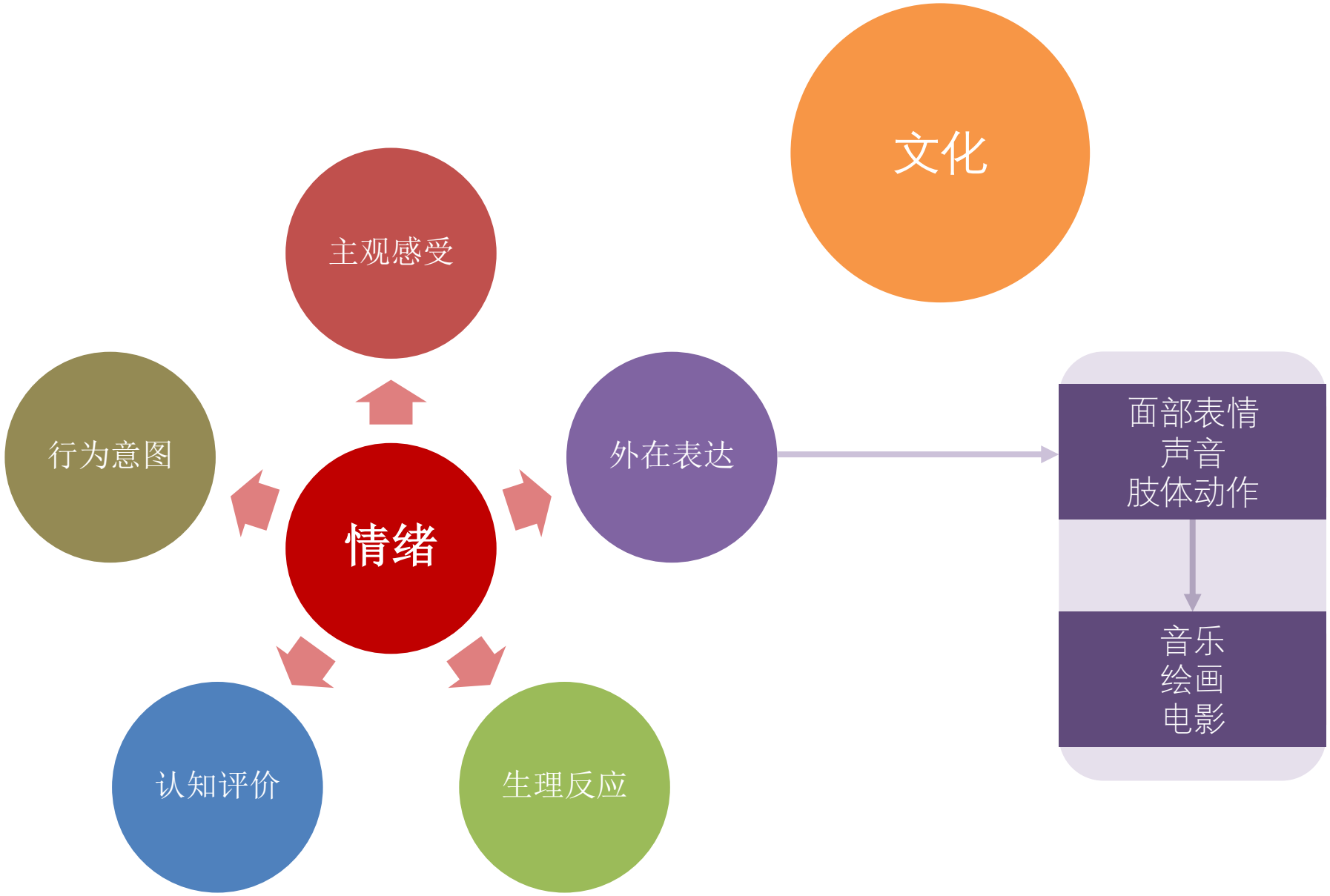
The question of universality

- Are there universal expressions of emotion?

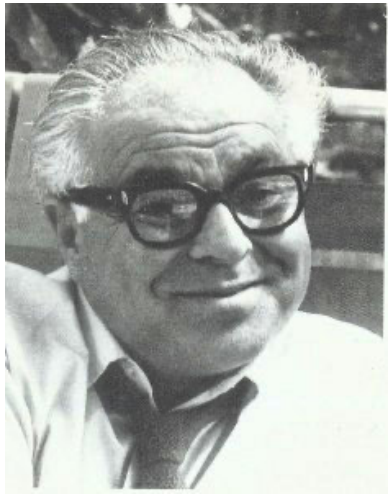


The question of universality




文化
主观感受
行为意图
情绪
外在表达
面部表情
声音
肢体动作
音乐
绘画
电影
认知评价
生理反应

A Brief History of Cross-Cultural Studies of Emotion







Tomkins, 1962/1963
Izard & Ekman, 1977/1992

Cultural Similarities

Evolutionary Theories of Emotion

Frijda, 1986
Ellsworth & Scherer, 2003
Russell & Barrett, 2003/2006

Cultural Differences

Appraisal and Constructionist Theories of Emotion

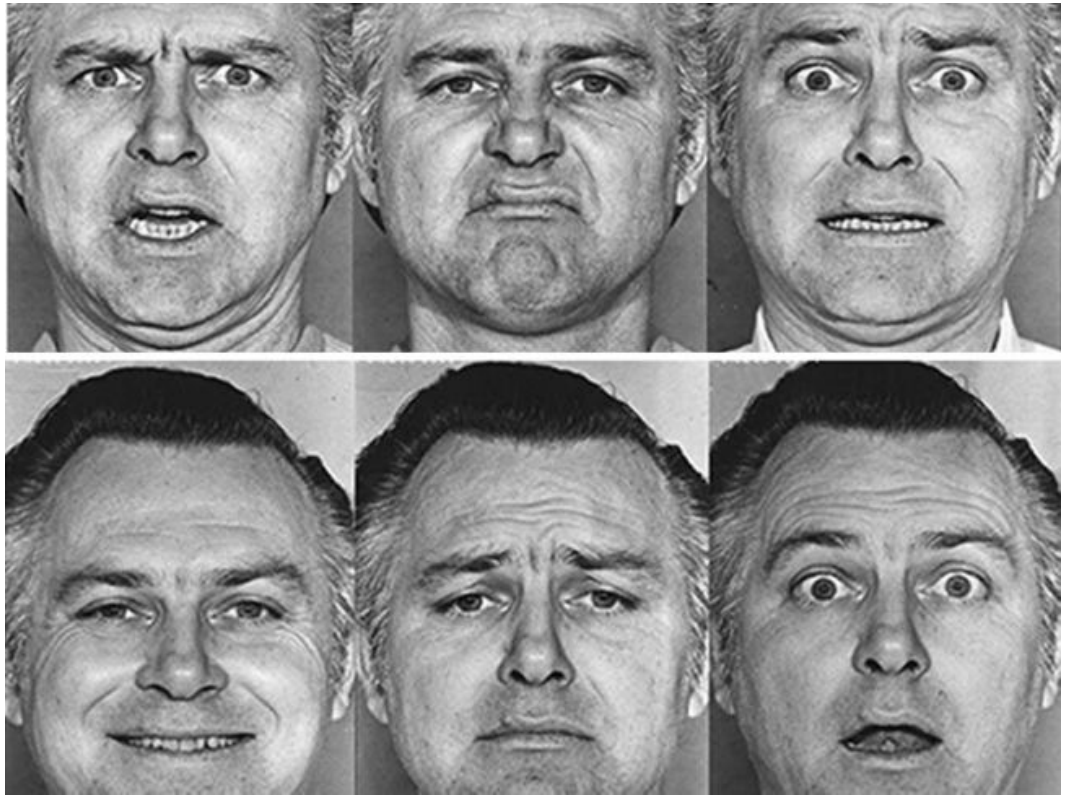
Elfenbein, 2002

Ingroup Advantage

Dialect Theories of Emotion

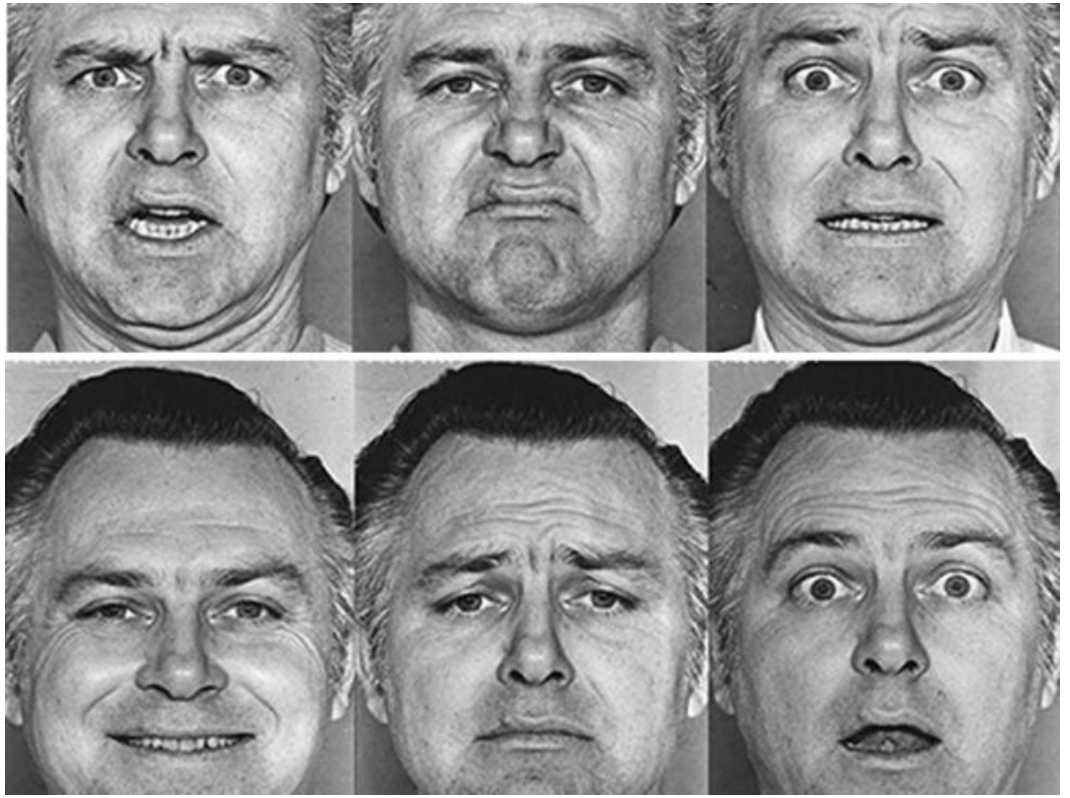
Universal facial expressions?

- Remote population
- Emotion scenarios
- Six emotions
- Bi-directional

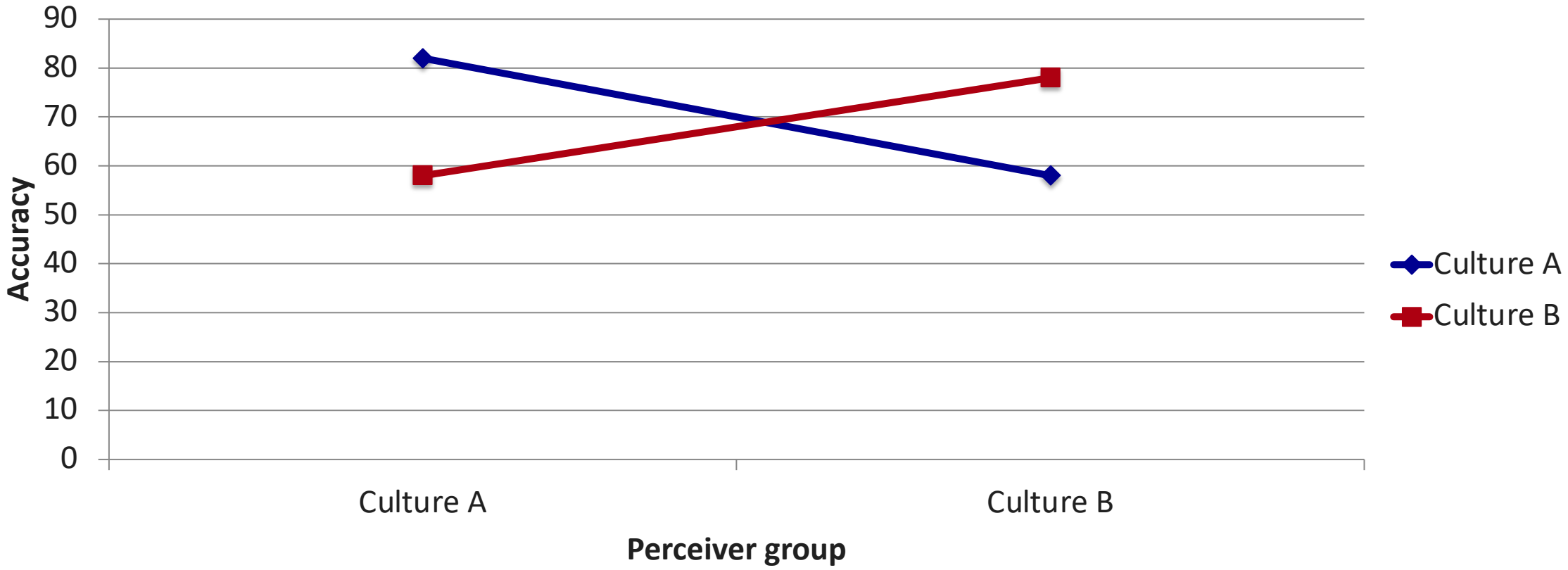


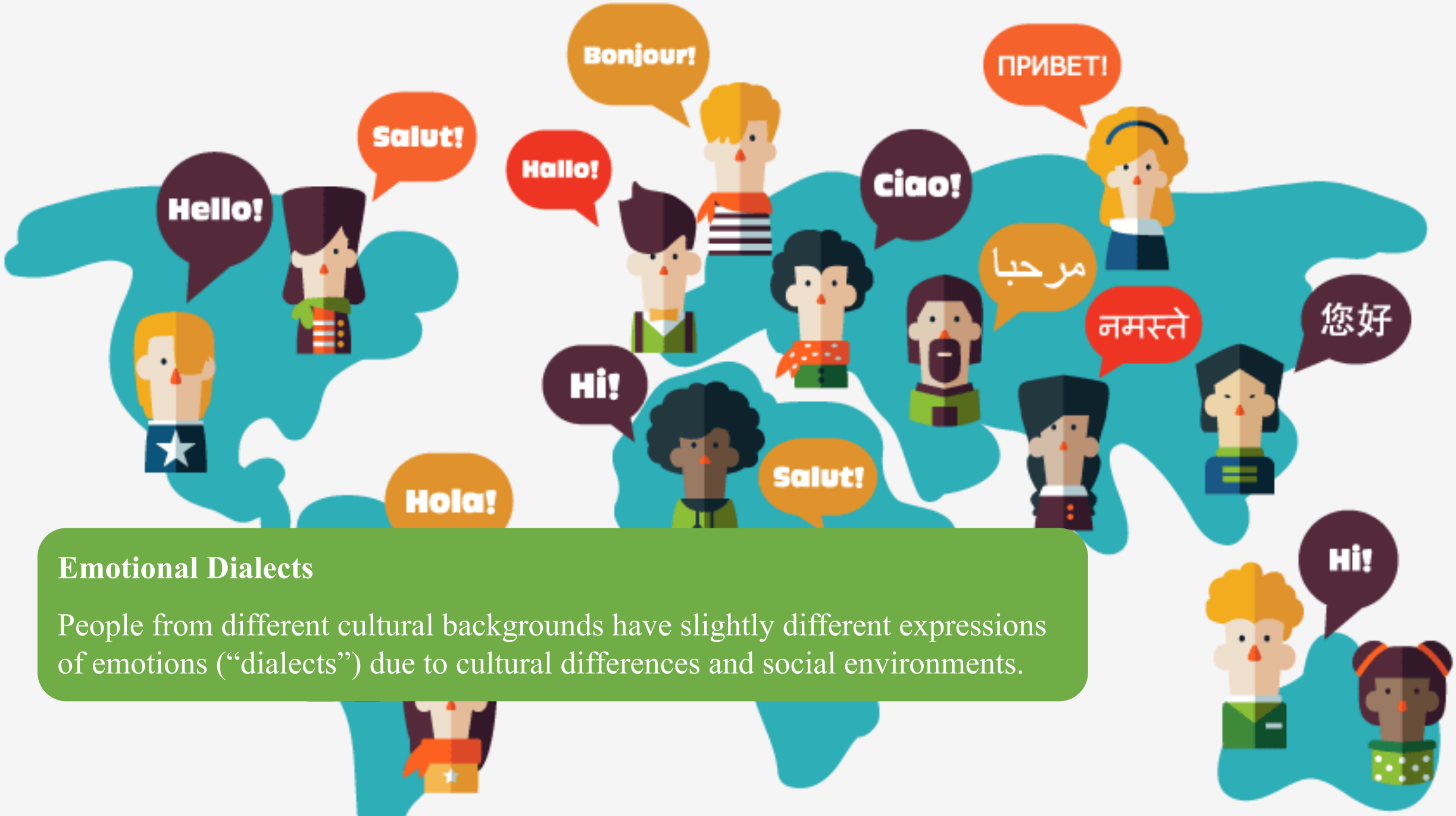
“Basic” facial expressions?

a) Anger
b) Disgust
c) Fear
d) Sadness
e) Surprise
f) Happiness



Ingroup advantage





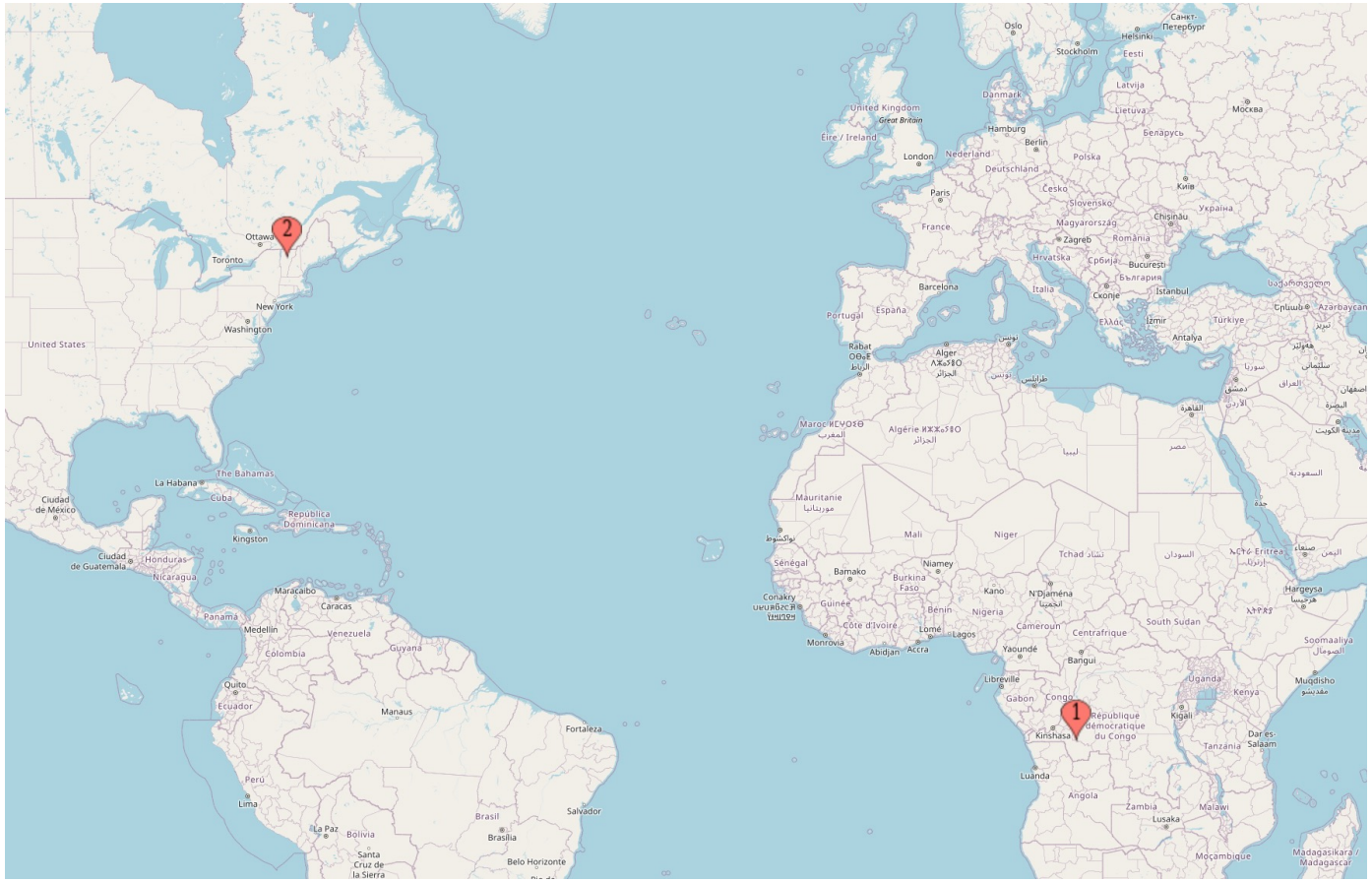
Emotional Dialects

People from different cultural backgrounds have slightly different expressions of emotions (“dialects”) due to cultural differences and social environments.

Emotional Dialects

Individuals are better at recognizing facial expressions of emotion from their own culture than from another culture.

Ingroup Advantage



Elfenbein et al. (2007), *Emotion*

Ingroup Advantage

Prototype



AU 1+4+15b

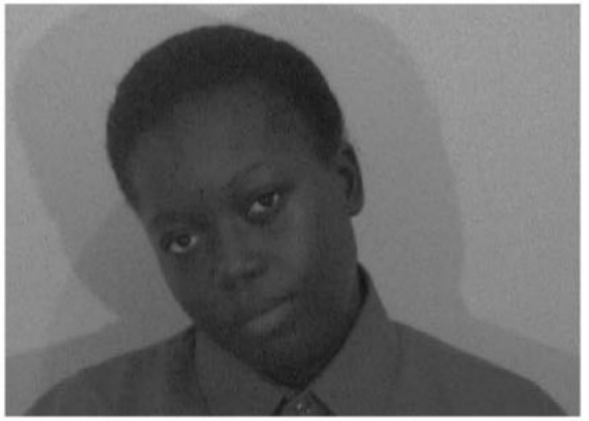
Quebecois

Sadness



AU 4

Gabonese



AU 56


在不同文化中都很相似。
are quite similar across cultures.



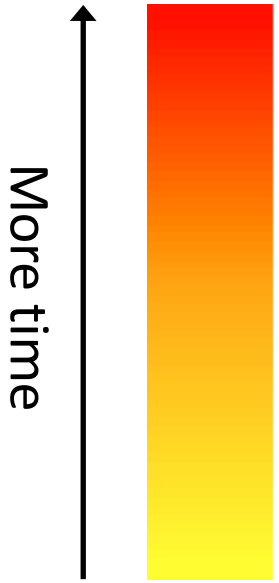


Concurrent Context



(Masuda et al., *JPSP*, 2008)

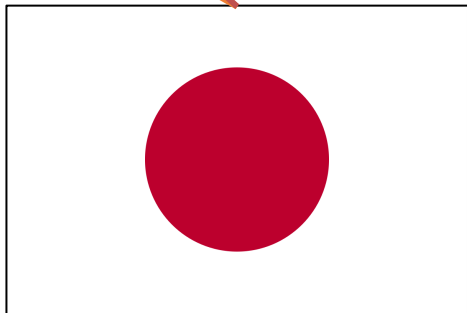
Concurrent Context





(Masuda et al., *JPSP*, 2008)





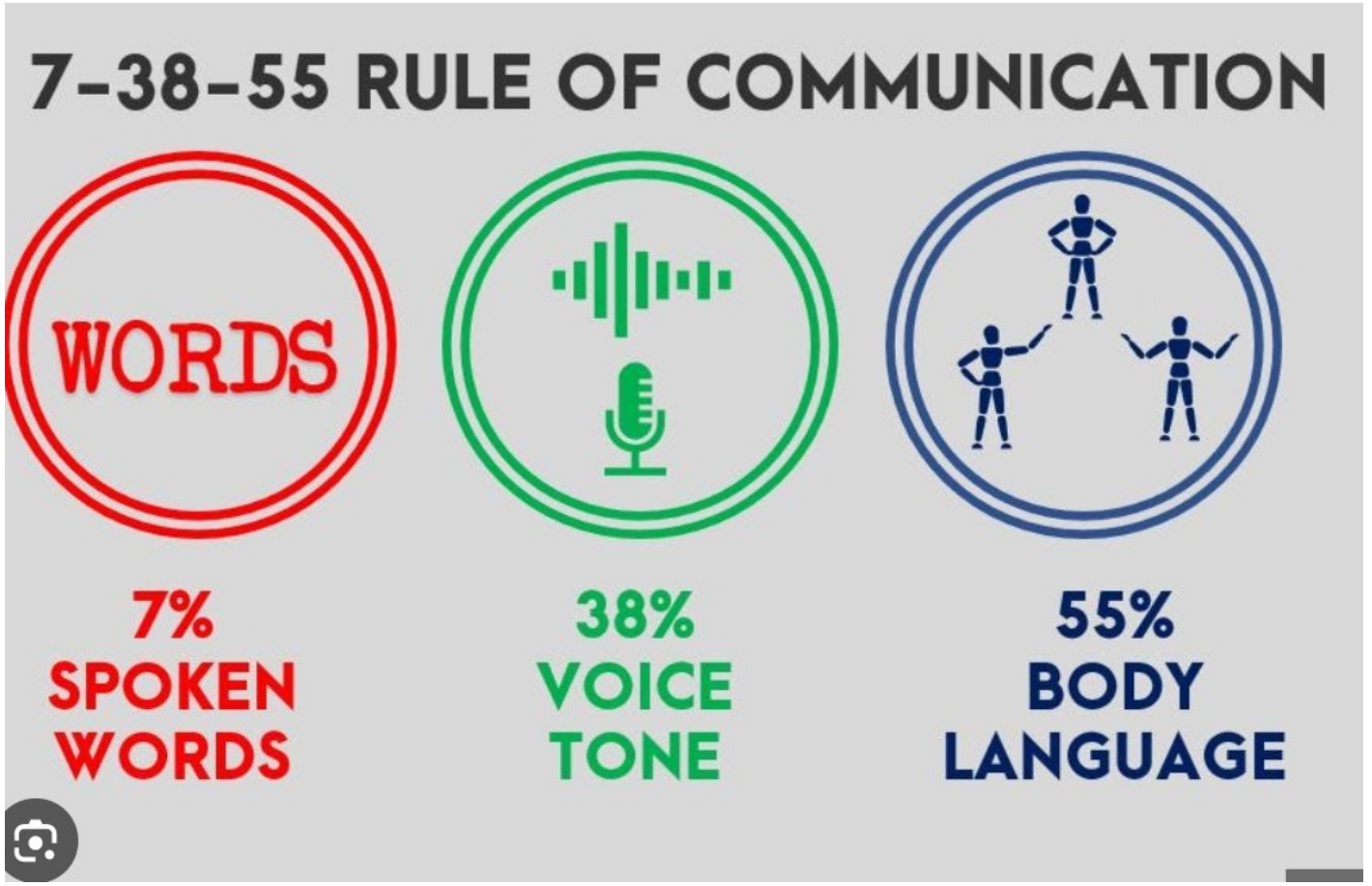
Concurrent Context





Compared to Westerners, Easterners are influenced more by the concurrent emotional context when judging others' facial expressions.

(Ito et al., *PSPB*, 2013)



Mehrabian A. & Wiener M (1967). Decoding of inconsistent communications. *JPSP, 6*(1).



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